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Fluid routing plug

Abstract

A fluid routing plug for use with a fluid end section. The fluid end section being one of a plurality of fluid end sections making up a fluid end side of a high pressure pump. The fluid routing plug is installed within a horizontal bore formed in a fluid end section and is configured to route fluid between an intake and discharge bore. The fluid routing plug comprises a plurality of first and second fluid passages. The first and second passages do not intersect and are offset from one another. The first fluid passages are configured to direct fluid delivered to the horizontal bore from intake bores towards a reciprocating plunger. The second fluid passages are configured to direct fluid pressurized by the plunger towards a discharge bore.

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Background/Summary

RELATED APPLICATIONS (1) This application is a Continuation-in-Part of U.S. Ser. No. 16/951,605, authored by Thomas et al. and filed on Nov. 18, 2020; and is a Continuation-in-Part of U.S. Ser. No. 17/550,552, authored by Thomas et al. and filed on Dec. 14, 2021. (2) This application also claims the benefit of the following provisional patent applications: Ser. No. 63/148,065, authored by Thomas et al. and filed on Feb. 10, 2021; Ser. No. 63/150,340, authored by Thomas et al. and filed on Feb. 17, 2021; Ser. No. 63/155,835, authored by Thomas et al. and filed on Mar. 3, 2021; Ser. No. 63/168,364, authored by Thomas et al. and filed on Mar. 31, 2021; and Ser. No. 63/283,487, authored by Thomas et al. and filed on Nov. 28, 2021. (3) The entire contents of all of the above listed non-provisional and provisional patent applications are incorporated herein by reference.

BACKGROUND

(1) Various industrial applications may require the delivery of high volumes of highly pressurized fluids. For example, hydraulic fracturing (commonly referred to as “fracking”) is a well stimulation technique used in oil and gas production, in which highly pressurized fluid is injected into a cased wellbore. As shown for example in FIG. 1, the pressured fluid flows through perforations **10** in a casing **12** and creates fractures **14** in deep rock formations **16**. Pressurized fluid is delivered to the casing **12** through a wellhead **18** supported on the ground surface **20**. Sand or other small particles (commonly referred to as “proppants”) are normally delivered with the fluid into the rock formations **16**. The proppants help hold the fractures **14** open after the fluid is withdrawn. The resulting fractures **14** facilitate the extraction of oil, gas, brine, or other fluid trapped within the rock formations **16**.

(2) Fluid ends are devices used in conjunction with a power source to pressurize the fluid used during hydraulic fracturing operations. A single fracking operation may require the use of two or more fluid ends at one time. For example, six fluid ends **22** are shown operating at a wellsite **24** in FIG. 2. Each of the fluid ends **22** is attached to a power end **26** in a one-to-one relationship. The power end **26** serves as an engine or motor for the fluid end **22**. Together, the fluid end **22** and power end **26** function as a hydraulic pump.

(3) Continuing with FIG. 2, a single fluid end **22** and its corresponding power end **26** are typically positioned on a truck bed **28** at the wellsite **24** so that they may be easily moved, as needed. The fluid and proppant mixture to be pressurized is normally held in large tanks **30** at the wellsite **24**.

An intake piping system **32** delivers the fluid and proppant mixture from the tanks **30** to each fluid end **22**. A discharge piping system **33** transfers the pressurized fluid from each fluid end **22** to the wellhead **18**, where it is delivered into the casing **12** shown in FIG. **1**.

(4) Fluid ends operate under notoriously extreme conditions, enduring the same pressures, vibrations, and abrasives that are needed to fracture the deep rock formations shown in FIG. **1**. Fluid ends may operate at pressures of 5,000-15,000 pounds per square inch (psi) or greater. Fluid used in hydraulic fracturing operations is typically pumped through the fluid end at a pressure of at least 8,000 psi, and more typically between 10,000 and 15,000 psi. However, the pressure may reach up to 22,500 psi. The power end used with the fluid end typically has a power output of at least 2,250 horsepower during hydraulic fracturing operations. A single fluid end typically produces a fluid volume of about 400 gallons, or 10 barrels, per minute during a fracking operation. A single fluid end may operate in flow ranges from 170 to 630 gallons per minute, or approximately 4 to 15 barrels per minute. When a plurality of fluid ends are used together, the fluid ends collectively deliver about 4,200 gallons per minute or 100 barrels per minute to the wellbore.

(5) In contrast, mud pumps known in the art typically operate at a pressure of less than 8,000 psi. Mud pumps are used to deliver drilling mud to a rotating drill bit within the wellbore during drilling operations. Thus, the drilling mud does not need to have as high of fluid pressure as fracking fluid. A fluid end does not pump drilling mud. A power end used with mud pumps typically has a power output of less than 2,250 horsepower. Mud pumps generally produce a fluid volume of about 150-600 gallons per minute, depending on the size of pump used.

(6) In further contrast, a fluid jetting pump known in the art typically operates at pressures of 30,000-90,000 psi. Jet pumps are used to deliver a highly concentrated stream of fluid to a desired area. Jet pumps typically deliver fluid through a wand. Fluid ends do not deliver fluid through a wand. Unlike fluid ends, jet pumps are not used in concert with a plurality of other jet pumps. Rather, only a single jet pump is used to pressurize fluid. A power end used with a jet pump typically has a power output of about 1,000 horsepower. Jet pumps generally produce a fluid volume of about 10 gallons per minute.

(7) High operational pressures may cause a fluid end to expand or crack. Such a structural failure may lead to fluid leakage, which leaves the fluid end unable to produce and maintain adequate fluid pressures. Moreover, if proppants are included in the pressurized fluid, those proppants may cause erosion at weak points within the fluid end, resulting in additional failures.

(8) It is not uncommon for conventional fluid ends to experience failure after only several hundred operating hours. Yet, a single fracking operation may require as many as fifty (50) hours of fluid end operation. Thus, a traditional fluid end may require replacement after use on as few as two fracking jobs.

(9) During operation of a hydraulic pump, the power end is not exposed to the same corrosive and abrasive fluids that move through the fluid end. Thus, power ends typically have much longer lifespans than fluid ends. A typical power end may service five or more different fluid ends during its lifespan.

(10) With reference to FIG. **3**, a traditional power end **34** is shown. The power end **34** comprises a housing **36** having a mounting plate **38** formed on its front end **40**. A plurality of stay rods **42** are attached to and project from the mounting plate **38**. A plurality of pony rods **44** are disposed at least partially within the power end **34** and project from openings formed in the mounting plate **38**. Each of the pony rods **44** is attached to a crank shaft installed within the housing **36**. Rotation of the crank shaft powers reciprocal motion of the pony rods **44** relative to the mounting plate **38**.

(11) A fluid end **46** shown in FIG. **3** is attached to the power end **34**. The fluid end **46** comprises a single housing **48** having a flange **50** machined therein. The flange **50** provides a connection point for the plurality of stay rods **42**. The stay rods **42** rigidly interconnect the power end **34** and the fluid end **46**. When connected, the fluid end **46** is suspended in offset relationship to the power end **34**.

(12) A plurality of plungers **52** are disposed within the fluid end **46** and project from openings formed in the flange **50**. The plungers **52** and pony rods **44** are arranged in a one-to-one relationship, with each plunger **52** aligned with and connected to a corresponding one of the pony rods **44**. Reciprocation of each pony rod **44** causes its connected plunger **52** to reciprocate within the fluid end **46**. In operation, reciprocation of the plungers **52** pressurizes fluid within the fluid end **46**. The reciprocation cycle of each plunger **52** is differently phased from that of each adjacent plunger **52**.

(13) With reference to FIG. 5, the interior of the fluid end **46** includes a plurality of longitudinally spaced bore pairs. Each bore pair includes a vertical bore **56** and an intersecting horizontal bore **58**. The zone of intersection between the paired bores defines an internal chamber **60**. Each plunger **52** extends through a horizontal bore **58** and into its associated internal chamber **60**. The plungers **52** and horizontal bores **58** are arranged in a one-to-one relationship.

(14) Each horizontal bore **58** is sized to receive a plurality of packing seals **64**. The seals **64** are configured to surround the installed plunger **52** and prevent high-pressure fluid from passing around the plunger **52** during operation. The packing seals **64** are maintained within the bore **58** by a retainer **65**. The retainer **65** has external threads **63** that mate with internal threads **67** formed in the walls surrounding the bore **58**. In some traditional fluid ends, the packing seals **64** are installed within a removable stuffing box sleeve that is installed within the horizontal bore.

(15) Each vertical bore **56** interconnects opposing top and bottom surfaces **66** and **68** of the fluid end **46**. Each horizontal bore **58** interconnects opposing front and rear surfaces **70** and **72** of the fluid end **46**. A discharge plug **74** seals each opening of each vertical bore **56** on the top surface **66** of the fluid end **46**. Likewise, a suction plug **76** seals each opening of each horizontal bore **58** on the front surface **70** of the fluid end **46**.

(16) Each of the plugs **74** and **76** features a generally cylindrical body. An annular seal **77** is installed within a recess formed in an outer surface of that body, and blocks passage of high pressure fluid. The discharge and suction plugs **74** and **76** are retained within their corresponding bores **56** and **58** by a retainer **78**, shown in FIGS. 3, 5, and 6. The retainer **78** has a cylindrical body having external threads **79** formed in its outer surface. The external threads **79** mate with internal threads **81** formed in the walls surrounding the bore **56** or **58** between the installed plug **74** or **76** and the surface **66** or **70** of the fluid end **46**.

(17) As shown in FIG. 3, a single manifold **80** is attached to the fluid end **46**. The manifold **80** is also connected to an intake piping system, of the type shown in FIG. 2. Fluid to be pressurized is drawn from the intake piping system into the manifold **80**, which directs the fluid into each of the vertical bores **56**, by way of openings (not shown) in the bottom surface **68**.

(18) When a plunger **52** is retracted, fluid is drawn into each internal chamber **60** from the manifold **80**. When a plunger **52** is extended, fluid within each internal chamber **60** is pressurized and forced towards a discharge conduit **82**. Pressurized fluid exits the fluid end **46** through one or more discharge openings **84**, shown in FIGS. 3-5. The discharge openings **84** are in fluid communication with the discharge conduit **82**. The discharge openings **84** are attached to a discharge piping system, of the type shown in FIG. 2.

(19) A pair of valves **86** and **88** are installed within each vertical bore **56**, on opposite sides of the internal chamber **60**. The valve **86** prevents backflow in the direction of the manifold **80**, while the valve **88** prevents backflow in the direction of the internal chamber **60**. The valves **86** and **88** each comprise a valve body **87** that seals against a valve seat **89**.

(20) Traditional fluid ends are normally machined from high strength alloy steel. Such material can corrode quickly, leading to fatigue cracks. Fatigue cracks occur because corrosion of the metal decreases the metal's fatigue strength—the amount of loading cycles that can be applied to a metal before it fails. Such cracking can allow leakage that prevents a fluid end from achieving and maintaining adequate pressures. Once such leakage occurs, fluid end repair or replacement becomes necessary.

(21) Fatigue cracks in fluid ends are commonly found in areas that experience high stress. For example, with reference to the fluid end **46** shown in FIG. 5, fatigue cracks are common at a corner **90** formed in the interior of the fluid end **46** by the intersection of the walls surrounding the horizontal bore **58** with the walls surrounding the vertical bore **56**. A plurality of the corners **90** surround each internal chamber **60**. Because fluid is pressurized within each internal chamber **60**, the corners **90** typically experience the highest amount of stress during operation, leading to fatigue cracks. Fatigue cracks are also common at the neck that connects the flange **50** and the housing **48**. Specifically, fatigue cracks tend to form at an area **92** where the neck joins the housing **48**, as shown for example in FIGS. 4 and 5.

(22) For the above reasons, there is a need in the industry for a fluid end configured to avoid or significantly delay the structures or conditions that cause wear or failures within a fluid end.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is an illustration of the underground environment of a hydraulic fracturing operation.
- (2) FIG. 2 illustrates above-ground equipment used in a hydraulic fracturing operation.
- (3) FIG. 3 is a left side perspective view of a traditional fluid end attached to a traditional power end.
- (4) FIG. 4 is a top plan view of the fluid end shown in FIG. 3.
- (5) FIG. 5 is a sectional view of the fluid end shown in FIG. 4, taken along line A-A.
- (6) FIG. 6 is a front perspective view of a fluid end. A plurality of stay rods are attached to the fluid end.
- (7) FIG. 7 is a rear perspective view of the fluid end shown in FIG. 6, but the plurality of stay rods have been removed.
- (8) FIG. 7A is a side elevational view of the fluid end shown in FIG. 6, but with another embodiment of intake and discharge manifolds.
- (9) FIG. 7B is a front perspective view of the fluid end shown in FIG. 7A.
- (10) FIG. 8 is a front perspective view of one of the fluid end sections making up the fluid end shown in FIG. 6.
- (11) FIG. 9 is a cross-sectional view of the fluid end section shown in FIG. 8, taken along line D-D.
- (12) FIG. 10 is a perspective view of a first surface of a connect plate used with the fluid end shown in FIG. 6.
- (13) FIG. 11 is a perspective view of a second surface of the connect plate shown in FIG. 10.
- (14) FIG. 12 is an elevational view of the first surface of the connect plate shown in FIG. 10.
- (15) FIG. 13 is an elevational view of the second surface of the connect plate shown in FIG. 10.
- (16) FIG. 14 is a perspective view of a second surface of a housing making up the fluid end section shown in FIG. 8.
- (17) FIG. 15 is an elevational view of the second surface of the housing shown in FIG. 14.
- (18) FIG. 16 is a cross-sectional view of the fluid end and stay rods shown in FIG. 6, taken along a plane that includes the line B-B.
- (19) FIG. 17 is a cross-sectional view of the fluid end and stay rods shown in FIG. 6, taken along a plane that includes the line C-C.
- (20) FIG. 18 is a top plan view of the housing shown in FIG. 14.
- (21) FIG. 19 is an enlarged view of area E shown in FIG. 9.
- (22) FIG. 20 is the cross-sectional view of the fluid end section shown in FIG. 17 with the upper and lower intake manifolds shown attached to the housing.
- (23) FIG. 21 is a rear perspective view of the fluid end section shown in FIG. 20, but the plunger has been removed.

- (24) FIG. 22 is a top plan view of a stuffing box shown attached to the housing in
- (25) FIG. 20.
- (26) FIG. 23 is a perspective view of a first surface of the stuffing box shown in FIG. 22.
- (27) FIG. 24 is an elevational view of the first surface of the stuffing box shown in FIG. 22.
- (28) FIG. 25 is a cross-sectional view of the stuffing box shown in FIG. 24, taken along line F-F.
- (29) FIG. 26 is a cross-sectional view of the stuffing box shown in FIG. 24, taken along line G-G.
- (30) FIG. 27 is a perspective view of a second surface of the stuffing box shown in FIG. 22.
- (31) FIG. 28 is an elevational view of the second surface of the stuffing box shown in FIG. 22.
- (32) FIG. 29 is a cross-sectional view of the stuffing box shown in FIG. 28, taken along line H-H.
- (33) FIG. 30 is a top plan view of a retainer shown attached to the stuffing box in FIG. 20.
- (34) FIG. 31 is a perspective view of a first surface of the retainer shown in FIG. 30.
- (35) FIG. 32 is an elevational view of the first surface of the retainer shown in FIG. 30.
- (36) FIG. 33 is a cross-sectional view of the retainer shown in FIG. 32, taken along line I-I.
- (37) FIG. 34 is a cross-sectional view of the retainer shown in FIG. 36, taken along line J-J.
- (38) FIG. 35 is a perspective view of a second surface of the retainer shown in FIG. 30.
- (39) FIG. 36 is an elevational view of the second surface of the retainer shown in FIG. 30.
- (40) FIG. 37 is a cross-sectional view of the retainer shown in FIG. 36, taken along line K-K.
- (41) FIG. 38 is a top plan view of a plunger packing shown installed within the stuffing box and retainer in FIG. 20.
- (42) FIG. 39 is a perspective view of a first surface of the plunger packing shown in FIG. 38.
- (43) FIG. 40 is an elevational view of the first surface of the plunger packing shown in FIG. 38.
- (44) FIG. 41 is a cross-sectional view of the plunger packing shown in FIG. 40, taken along line L-L.
- (45) FIG. 42 is a perspective exploded view of the plunger packing shown in FIG. 38.
- (46) FIG. 43 is a top plan view of a packing nut shown installed within the retainer in FIG. 20.
- (47) FIG. 44 is a perspective view of a first surface of the packing nut shown in FIG. 43.
- (48) FIG. 45 is an elevational view of the first surface of the packing nut shown in FIG. 43.
- (49) FIG. 46 is a cross-sectional view of the packing nut shown in FIG. 45, taken along line M-M.
- (50) FIG. 47 is a perspective view of a first surface of a retainer shown installed within the housing in FIG. 20.
- (51) FIG. 48 is an elevational view of the first surface of the retainer shown in FIG. 47.
- (52) FIG. 49 is a cross-sectional view of the retainer shown in FIG. 48, taken along line N-N.
- (53) FIG. 50 is the cross-sectional view shown in FIG. 9, but the suction valve is spaced from the fluid routing plug.
- (54) FIG. 51 is the cross-sectional view shown in FIG. 50, but the plunger has extended into the housing, the suction valve is sealed against the fluid routing plug, and the discharge valve is spaced from the fluid routing plug.
- (55) FIG. 52 is a perspective view of a second surface of a fluid routing plug shown installed within the fluid end section in FIG. 50.
- (56) FIG. 53 is a perspective view of a first surface of the fluid routing plug shown in FIG. 52.
- (57) FIG. 54 is an elevational view of the second surface of the fluid routing plug shown in FIG. 52.
- (58) FIG. 55 is a cross-sectional view of the fluid routing plug shown in FIG. 54, taken along line O-O.
- (59) FIG. 56 is an elevational view of the first surface of the fluid routing plug shown in FIG. 52.
- (60) FIG. 57 is a top plan view of the fluid routing plug shown in FIG. 52.
- (61) FIG. 58 is a cross-sectional view of the fluid routing plug shown in FIG. 57, taken along line P-P.
- (62) FIG. 59 is an enlarged view of area Q shown in FIG. 57.
- (63) FIG. 60 is the top plan view of the fluid routing plug shown in FIG. 57, but the plug has been

slightly rotated.

(64) FIG. **61** is a cross-sectional view of the fluid routing plug shown in FIG. **60**, taken along line R-R.

(65) FIG. **62** is a cross-sectional view of the fluid routing plug shown in FIG. **60**, taken along line S-S.

(66) FIG. **63** is a cross-sectional view of the fluid routing plug shown in FIG. **60**, taken along line T-T.

(67) FIG. **64** is an enlarged view of the fluid routing plug shown in FIG. **60**.

(68) FIG. **65** is the cross-sectional view shown in FIG. **50**.

(69) FIG. **66** is an enlarged view of area U shown in FIG. **65**.

(70) FIG. **67** is an enlarged view of area V shown in FIG. **65**.

(71) FIG. **68** is an enlarged view of area W shown in FIG. **65**.

(72) FIG. **69** is an enlarged view of area X shown in FIG. **65**.

(73) FIG. **70** is an enlarged view of area Y shown in FIG. **65**.

(74) FIG. **71** is an enlarged view of area Z shown in FIG. **65**.

(75) FIG. **72** is a top plan view of a suction valve shown installed within the housing in FIG. **50**.

(76) FIG. **73** is a perspective view of a second surface of the suction valve shown in FIG. **72**.

(77) FIG. **74** is an elevational view of the second surface of the suction valve shown in FIG. **72**.

(78) FIG. **75** is a perspective view of a first surface of the suction valve shown in FIG. **72**.

(79) FIG. **76** is a cross-sectional view of the suction valve shown in FIG. **74**, taken along line AA-AA.

(80) FIG. **77** is a top plan view of a suction valve guide shown installed within the housing shown in FIG. **50**.

(81) FIG. **78** is a perspective view of a first surface of the suction valve guide shown in FIG. **77**.

(82) FIG. **79** is an elevation view of the first surface of the suction valve guide shown in FIG. **77**.

(83) FIG. **80** is a cross-sectional view of the suction valve guide shown in FIG. **79**, taken along line AB-AB.

(84) FIG. **81** is a perspective view of a second surface of the suction valve guide shown in FIG. **77**.

(85) FIG. **82** is an elevational view of the second surface of the suction valve guide shown in FIG. **77**.

(86) FIG. **83** is a perspective view of the suction valve guide shown in FIG. **77** engaged with the suction valve shown in FIG. **72**. A spring is shown positioned between the suction valve guide and the suction valve.

(87) FIG. **84** is a top plan view of the suction valve guide, suction valve, and spring shown in FIG. **83**.

(88) FIG. **85** is a top plan view of a discharge valve shown installed within the housing in FIG. **50**.

(89) FIG. **86** is a perspective view of a second surface of the discharge valve shown in FIG. **85**.

(90) FIG. **87** is an elevational view of the second surface of the discharge valve shown in FIG. **85**.

(91) FIG. **88** is a perspective view of a first surface of the discharge valve shown in FIG. **85**.

(92) FIG. **89** is a cross-sectional view of the discharge valve shown in FIG. **87**, taken along line AC-AC.

(93) FIG. **90** is a top plan view of a discharge valve guide shown installed within the housing in FIG. **50**.

(94) FIG. **91** is a perspective view of a first surface of the discharge valve guide shown in FIG. **90**.

(95) FIG. **92** is an elevation view of the first surface of the discharge valve guide shown in FIG. **90**.

(96) FIG. **93** is a cross-sectional view of the discharge valve guide shown in FIG. **92**, taken along line AD-AD.

(97) FIG. **94** is a cross-sectional view of the discharge valve guide shown in FIG. **92**, taken along line AE-AE.

(98) FIG. **95** is a perspective view of a second surface of the discharge valve guide shown in FIG.

- 90.**
- (99) FIG. **96** is a perspective cut-away view of a first surface of the fluid end section shown in FIG. **8**.
- (100) FIG. **97** is an enlarged view of area AF shown in FIG. **96**.
- (101) FIG. **98** is a perspective view of the discharge valve guide shown in FIG. **90** engaged with the discharge valve shown in FIG. **85**. A spring is shown positioned between the discharge valve guide and the discharge valve.
- (102) FIG. **99** is a top plan view of the discharge valve guide, discharge valve, and spring shown in FIG. **98**.
- (103) FIG. **100** is the cross-sectional view of the fluid end section shown in FIG. **9**.
- (104) FIG. **100A** is a perspective view of a second surface of another embodiment of a fluid routing plug.
- (105) FIG. **100B** is a perspective view of a first surface of the fluid routing plug shown in FIG. **100A**.
- (106) FIG. **100C** is an elevational view of the second surface of the fluid routing plug shown in FIG. **100A**.
- (107) FIG. **100D** is a cross-sectional view of the fluid routing plug shown in FIG. **100C**, taken along line JA-JA.
- (108) FIG. **100E** is a cross-sectional view of the fluid routing plug shown in FIG. **100C**, taken along line JB-JB.
- (109) FIG. **100F** is the cross-sectional view of the fluid end section shown in FIG. **9**, but the fluid routing plug from FIG. **100A** is shown installed within the housing.
- (110) FIG. **100G** is an enlarged view of area JC from FIG. **100F**.
- (111) FIG. **101** is a perspective view of a first surface of another embodiment of a fluid routing plug.
- (112) FIG. **102** is an elevational view of the first surface of the fluid routing plug shown in FIG. **101**.
- (113) FIG. **103** is a cross-sectional view of the fluid routing plug shown in FIG. **102**, taken along line AG-AG.
- (114) FIG. **104** is a top plan view of the fluid routing plug shown in FIG. **101**.
- (115) FIG. **105** is a perspective view of a second surface of the fluid routing plug shown in FIG. **101**.
- (116) FIG. **106** is an elevational view of the second surface of the fluid routing plug shown in FIG. **101**.
- (117) FIG. **107** is a cross-sectional view of the fluid routing plug shown in FIG. **106**, taken along line AH-AH.
- (118) FIG. **108** is the cross-sectional view of the fluid end section shown in FIG. **50**, but the fluid routing plug from FIG. **101** is shown installed within the housing.
- (119) FIG. **109** is the cross-sectional view of the fluid end section shown in FIG. **51**, but the fluid routing plug from FIG. **101** is shown installed within the housing.
- (120) FIG. **110** is a top plan view of another embodiment of a suction and discharge valve.
- (121) FIG. **111** is a perspective view of a second surface of the suction and discharge valve shown in FIG. **110**.
- (122) FIG. **112** is an elevational view of a second surface of the suction and discharge valve shown in FIG. **110**.
- (123) FIG. **113** is a perspective view of a first surface of the suction and discharge valve shown in FIG. **110**.
- (124) FIG. **114** is a cross-sectional view of the suction and discharge valve shown in FIG. **112**, taken along line AI-AI.
- (125) FIG. **115** is the cross-sectional view of the fluid end section shown in FIG. **65**, but another

embodiment of a fluid routing plug is shown installed within the housing.

(126) FIG. **116** is an enlarged view of area AJ shown in FIG. **115**.

(127) FIG. **117** is an enlarged view of area AK shown in FIG. **115**.

(128) FIG. **118** is the cross-sectional view of the fluid end section shown in FIG. **65**, but another embodiment of a fluid routing plug is shown installed within the housing.

(129) FIG. **119** is an enlarged view of area AL shown in FIG. **118**.

(130) FIG. **120** is an enlarged view of area AM shown in FIG. **118**.

(131) FIG. **121** is a top plan view of another embodiment of a fluid routing plug.

(132) FIG. **122** is a perspective view of a second surface of the fluid routing plug shown in FIG. **121**, with a plurality of second fluid passages formed within the plug shown by phantom lines.

(133) FIG. **123** is an elevational view of the second surface of the fluid routing plug shown in FIG. **121**, with a plurality of second fluid passages formed within the plug shown by phantom lines.

(134) FIG. **124** is a cross-sectional view of the fluid routing plug shown in FIG. **123**, taken along line AN-AN.

(135) FIG. **125** is a cross-sectional view of the fluid routing plug shown in FIG. **121**, taken along line AO-AO.

(136) FIG. **126** is the top plan view of the fluid routing plug shown in FIG. **121**, with the plurality of second fluid passages formed within the plug shown by phantom lines.

(137) FIG. **127** is an elevational view of a first surface of the fluid routing plug shown in FIG. **121**, with a plurality of second fluid passages formed within the plug shown by phantom lines.

(138) FIG. **128** is a perspective view of the first surface of the fluid routing plug shown in FIG. **121**.

(139) FIG. **128A** is a perspective view of a second surface of another embodiment of a fluid routing plug.

(140) FIG. **128B** is an elevational view of the second surface of the fluid routing plug shown in FIG. **128A**.

(141) FIG. **128C** is a cross-sectional view of the fluid routing plug shown in FIG. **128A**, taken along line KA-KA.

(142) FIG. **128D** is a top plan view of the fluid routing plug shown in FIG. **128A**.

(143) FIG. **128E** is a perspective view of a first surface of the fluid routing plug shown in FIG. **128A**.

(144) FIG. **128F** is an elevational view of the first surface of the fluid routing plug shown in FIG. **128A**.

(145) FIG. **128G** is a cross-sectional view of the fluid routing plug shown in FIG. **128D**, taken along line KB-KB.

(146) FIG. **129** is a top plan view of another embodiment of a fluid routing plug.

(147) FIG. **130** is an elevational view of a second surface of the fluid routing plug shown in FIG. **129**.

(148) FIG. **131** is a cross-sectional view of the fluid routing plug shown in FIG. **130**, taken along line AP-AP.

(149) FIG. **132** is a perspective view of a second surface of another embodiment of a fluid routing plug.

(150) FIG. **133** is a perspective view of a first surface of the fluid routing plug shown in FIG. **132**.

(151) FIG. **134** is an elevational view of the second surface of the fluid routing plug shown in FIG. **132**.

(152) FIG. **135** is a cross-sectional view of the fluid routing plug shown in FIG. **134**, taken along line AQ-AQ.

(153) FIG. **136** is a cross-sectional view of a fluid end section comprising another embodiment of a housing. The fluid routing plug shown in FIG. **132** is installed within the housing.

(154) FIG. **137** is an enlarged view of area AR shown in FIG. **136**.

(155) FIG. **138** is an enlarged view of area AS shown in FIG. **136**.

(156) FIG. **139** is a perspective view of a second surface of another embodiment of a fluid routing plug.

(157) FIG. **140** is a perspective view of a first surface of the fluid routing plug shown in FIG. **139**.

(158) FIG. **141** is an elevational view of the second surface of the fluid routing plug shown in FIG. **139**.

(159) FIG. **142** is a cross-sectional view of the fluid routing plug shown in FIG. **141**, taken along line AT-AT.

(160) FIG. **143** is a perspective view of a second surface of another embodiment of a fluid routing plug.

(161) FIG. **144** is a perspective view of a first surface of the fluid routing plug shown in FIG. **143**.

(162) FIG. **145** is an elevational view of the second surface of the fluid routing plug shown in FIG. **143**.

(163) FIG. **146** is a cross-sectional view of the fluid routing plug shown in FIG. **145**, taken along line AU-AU.

(164) FIG. **147** is a cross-sectional view of the fluid routing plug shown in FIG. **145**, taken along line AV-AV.

(165) FIG. **148** is an exploded perspective view of the second surface of the fluid routing plug shown in FIG. **143**.

(166) FIG. **149** is an exploded perspective view of the first surface of the fluid routing plug shown in FIG. **143**.

(167) FIG. **150** is a cross-sectional view of a fluid end section comprising another embodiment of a housing. The fluid routing plug shown in FIG. **143** is installed within the housing.

(168) FIG. **151** is a perspective view of a first surface of another embodiment of a fluid routing plug.

(169) FIG. **152** is a perspective view of a second surface of the fluid routing plug shown in FIG. **151**.

(170) FIG. **153** is an elevational view of the first surface of the fluid routing plug shown in FIG. **151**.

(171) FIG. **154** is a top plan view of the fluid routing plug shown in FIG. **151**.

(172) FIG. **155** is a cross-sectional view of the fluid routing plug shown in FIG. **153**, taken along line AW-AW.

(173) FIG. **156** is a cross-sectional view of the fluid routing plug shown in FIG. **154**, taken along line AX-AX.

(174) FIG. **157** is an exploded perspective view of the second surface of the fluid routing plug shown in FIG. **151**.

(175) FIG. **158** is an exploded perspective view of the first surface of the fluid routing plug shown in FIG. **151**.

(176) FIG. **159** is a cross-sectional view of another embodiment of a housing. The fluid routing plug shown in FIG. **151** is installed within the housing.

(177) FIG. **160** is a cross-sectional view of the housing shown in FIG. **159**, taken along a different axis.

(178) FIG. **161** is a top plan view of another embodiment of a fluid routing plug.

(179) FIG. **162** is an elevational view of a first surface of the fluid routing plug shown in FIG. **161**.

(180) FIG. **163** is a cross-sectional view of the fluid routing plug shown in FIG. **162**, taken along line AY-AY.

(181) FIG. **164** is a cross-sectional view of another embodiment of a housing. The fluid routing plug shown in FIG. **161** is installed within the housing.

(182) FIG. **165** is a perspective view of a first surface of another embodiment of a fluid routing plug.

(183) FIG. **166** is a top plan view of the fluid routing plug shown in FIG. **165**.

(184) FIG. **167** is an elevational view of the first surface of the fluid routing plug shown in FIG. **165**.

(185) FIG. **168** is a cross-sectional view of the fluid routing plug shown in FIG. **167**, taken along line AZ-AZ.

(186) FIG. **169** is an exploded perspective view of the first surface of the fluid routing plug shown in FIG. **165**.

(187) FIG. **170** is a cross-sectional view of another embodiment of a housing. The fluid routing plug shown in FIG. **165** is installed within the housing.

(188) FIG. **171** is a perspective view of a second surface of a first section of another embodiment of a fluid routing plug.

(189) FIG. **172** is an elevational view of the second surface of the first section shown in FIG. **171**.

(190) FIG. **173** is a cross-sectional view of the first section shown in FIG. **172**, taken along line BA-BA.

(191) FIG. **174** is a cross-sectional view of the first section shown in FIG. **172**, taken along line BB-BB.

(192) FIG. **175** is a perspective view of a first surface of the first section shown in FIG. **171**.

(193) FIG. **176** is a top plan view of the first section shown in FIG. **171**.

(194) FIG. **177** is an elevational view of the first surface of the first section shown in FIG. **171**.

(195) FIG. **178** is cross-sectional view of the first section shown in FIG. **177**, taken along line BC-BC.

(196) FIG. **179** is the cross-sectional view of the first section shown in FIG. **178**, but with an annular insert installed within the first surface.

(197) FIG. **180** is an exploded perspective view of the first surface of the first section shown in FIG. **171**.

(198) FIG. **181** is a perspective view of a first surface of a second section for use with the first section shown in FIG. **171**.

(199) FIG. **182** is a perspective view of the second surface of the second section shown in FIG. **181**.

(200) FIG. **183** is an elevational view of the second surface of the second section shown in FIG. **181**.

(201) FIG. **184** is a cross-sectional view of the second section shown in FIG. **183**, taken along line BD-BD.

(202) FIG. **185** is a cross-sectional view of another embodiment of a housing. The fluid routing plug made of the first section shown in FIG. **171** and the second section shown in FIG. **181** is installed within the housing.

(203) FIG. **186** is an enlarged view of area BE shown in FIG. **185**.

(204) FIG. **187** is a cross-sectional view of the housing shown in FIG. **185**, but taken along a different axis.

(205) FIG. **188** is a perspective view of a second surface of another embodiment of a first section.

(206) FIG. **189** is an elevational view of the second surface of the first section shown in FIG. **188**.

(207) FIG. **190** is a cross-sectional view of the first section shown in FIG. **189**, taken along line BF-BF.

(208) FIG. **191** is a cross-sectional view of the first section shown in FIG. **189**, taken along line BG-BG.

(209) FIG. **192** is a perspective view of a first surface of the first section shown in FIG. **188**.

(210) FIG. **193** is an elevational view of the first surface of the first section shown in FIG. **188**.

(211) FIG. **194** is a top plan view of the first section shown in FIG. **188**.

(212) FIG. **195** is a cross-sectional view of another embodiment of a housing.

(213) Another embodiment of a fluid routing plug made of the first section shown in FIG. **188** and

the second section shown in FIG. **181** is installed within the housing.

(214) FIG. **196** is a perspective view of a first surface of another embodiment of a first section.

(215) FIG. **197** is a perspective view of a second surface of the first section shown in FIG. **196**.

(216) FIG. **198** is an elevational view of the first surface of the first section shown in FIG. **196**.

(217) FIG. **199** is a top plan view of the first section shown in FIG. **196**.

(218) FIG. **200** is a cross-sectional view of the first section shown in FIG. **198**, taken along line BH-BH.

(219) FIG. **201** is a cross-sectional view of the first section shown in FIG. **199**, taken along line BI-BI.

(220) FIG. **202** is a cross-sectional view of another embodiment of a housing.

(221) Another embodiment of a fluid routing plug made of the first section shown in FIG. **196** and another embodiment of a second section is installed within the housing.

(222) FIG. **203** is a cross-sectional view of the housing shown in FIG. **202**, but taken along a different axis.

(223) FIG. **204** is a perspective view of a first surface of another embodiment of a first section.

(224) FIG. **205** is a perspective view of a second surface of the first section shown in FIG. **204**.

(225) FIG. **206** is an elevational view of the first surface of the first section shown in FIG. **204**.

(226) FIG. **207** is a top plan view of the first section shown in FIG. **204**.

(227) FIG. **208** is a cross-sectional view of the first section shown in FIG. **206**, taken along line BJ-BJ.

(228) FIG. **209** is a cross-sectional view of the first section shown in FIG. **207**, taken along line BK-BK.

(229) FIG. **210** is a cross-sectional view of another embodiment of a housing.

(230) Another embodiment of a fluid routing plug made of the first section shown in FIG. **204** and the second section shown in FIG. **202** is installed within the housing.

(231) FIG. **211** is a cross-sectional view of the housing shown in FIG. **210**, but taken along a different axis.

(232) FIG. **212** is a perspective view of the first surface of another embodiment of a fluid routing plug.

(233) FIG. **213** is a cross-sectional view of the fluid routing plug shown in FIG. **214**, taken along line BL-BL.

(234) FIG. **214** is an elevational view of the first surface of the fluid routing plug shown in FIG. **212**.

(235) FIG. **215** is a cross-sectional view of the fluid routing plug shown in FIG. **214**, taken along line BM-BM.

(236) FIG. **216** is a top plan view of the fluid routing plug shown in FIG. **212**.

(237) FIG. **217** is a perspective view of the second surface of the fluid routing plug shown in FIG. **212**.

(238) FIG. **218** is an elevational view of the second surface of the fluid routing plug shown in FIG. **212**.

(239) FIG. **219** is a cross-sectional view of the fluid routing plug shown in FIG. **216**, taken along line BN-BN.

(240) FIG. **220** is the cross-sectional view of the fluid routing plug shown in FIG. **213** having a pair of inserts installed therein.

(241) FIG. **221** is a perspective and exploded view of the first surface of the fluid routing plug shown in FIG. **220**.

(242) FIG. **222** is a perspective and exploded view of the second surface of the fluid routing plug shown in FIG. **220**.

(243) FIG. **223** is a perspective view of a first surface of another embodiment of a fluid routing plug.

(244) FIG. **224** is a cross-sectional view of the fluid routing plug shown in FIG. **225**, taken along line BO-BO.

(245) FIG. **225** is an elevational view of the first surface of the fluid routing plug shown in FIG. **223**.

(246) FIG. **226** is a top plan view of the fluid routing plug shown in FIG. **223**.

(247) FIG. **227** is a cross-sectional view of the fluid routing plug shown in FIG. **226**, taken along line BP-BP.

(248) FIG. **228** is the top plan view of the fluid routing plug shown in FIG. **226** having a pair of seals installed therein.

(249) FIG. **229** is the cross-sectional view of the fluid routing plug shown in FIG. **225**, taken along line BO-BO and having a pair of seals installed therein.

(250) FIG. **230** is a cross-sectional view of an alternative embodiment of a fluid end section having the fluid routing plug shown in FIG. **229** installed therein.

(251) FIG. **231** is a cross-sectional view of another embodiment of a fluid routing plug.

(252) FIG. **232** is a perspective view of the second surface of another embodiment of a fluid routing plug.

(253) FIG. **233** is a perspective view of the first surface of the fluid routing plug shown in FIG. **232**.

(254) FIG. **234** is a cross-sectional view of the fluid routing plug shown in FIG. **232**.

(255) FIG. **235** is a cross-sectional view of an alternative embodiment of a fluid end section having the fluid routing plug shown in FIG. **232** installed therein.

(256) FIG. **236** is an enlarged view of area BQ shown in FIG. **235**.

(257) FIG. **237** is a top plan view of another embodiment of a fluid routing plug.

(258) FIG. **238** is a perspective view of the second surface of the fluid routing plug shown in FIG. **237**.

(259) FIG. **239** is a side elevational view of the second surface of the fluid routing plug shown in FIG. **237**.

(260) FIG. **240** is a cross-sectional view of the fluid routing plug shown in FIG. **239**, taken along line BU-BU.

(261) FIG. **241** is a perspective view of the first surface of the fluid routing plug shown in FIG. **237**.

(262) FIG. **242** is a side elevational view of the first surface of the fluid routing plug shown in FIG. **237**.

(263) FIG. **243** is a cross-sectional view of the fluid routing plug shown in FIG. **237**, taken along line BR-BR.

(264) FIG. **244** is a cross-sectional view of the fluid routing plug shown in FIG. **237**, taken along line BS-BS.

(265) FIG. **245** is a cross-sectional view of the fluid routing plug shown in FIG. **237**, taken along line BT-BT.

(266) FIG. **246** is a side elevational and conical-section view of the second surface of the fluid routing plug shown in FIG. **237**, the conical-section is taken from circle BV shown in FIG. **238** to circle BW shown in FIG. **241**.

(267) FIG. **247** is top perspective view of the second surface of the fluid routing plug shown in FIG. **246**.

(268) FIG. **248** is a perspective view of the second surface of the fluid routing plug shown in FIG. **246**.

(269) FIG. **249** is a top plan view of another embodiment of a fluid routing plug.

(270) FIG. **250** is a perspective view of the second surface of the fluid routing plug shown in FIG. **249**.

(271) FIG. **251** is a side elevational view of the second surface of the fluid routing plug shown in

FIG. 249.

(272) **FIG. 252** is a cross-sectional view of the fluid routing plug shown in **FIG. 251**, taken along line CA-CA.

(273) **FIG. 253** is a perspective view of the first surface of the fluid routing plug shown in **FIG. 249**.

(274) **FIG. 254** is a side elevational view of the first surface of the fluid routing plug shown in **FIG. 249**.

(275) **FIG. 255** is a cross-sectional view of the fluid routing plug shown in **FIG. 249**, taken along line BX-BX.

(276) **FIG. 256** is a cross-sectional view of the fluid routing plug shown in **FIG. 249**, taken along line BY-BY.

(277) **FIG. 257** is a cross-sectional view of the fluid routing plug shown in **FIG. 249**, taken along line BZ-BZ.

(278) **FIG. 258** is a side elevational and conical-section view of the second surface of the fluid routing plug shown in **FIG. 249**, the conical-section is taken from circle CB shown in **FIG. 250** to circle CC shown in **FIG. 253**.

(279) **FIG. 259** is top perspective view of the second surface of the fluid routing plug shown in **FIG. 258**.

(280) **FIG. 260** is a perspective view of the second surface of the fluid routing plug shown in **FIG. 258**.

(281) **FIG. 261** is a top plan view of another embodiment of a fluid routing plug.

(282) **FIG. 262** is a perspective view of the second surface of the fluid routing plug shown in **FIG. 261**.

(283) **FIG. 263** is a side elevational view of the second surface of the fluid routing plug shown in **FIG. 261**.

(284) **FIG. 264** is a cross-sectional view of the fluid routing plug shown in **FIG. 263**, taken along line CH-CH.

(285) **FIG. 265** is a perspective view of the first surface of the fluid routing plug shown in **FIG. 261**.

(286) **FIG. 266** is a side elevational view of the first surface of the fluid routing plug shown in **FIG. 261**.

(287) **FIG. 267** is a cross-sectional view of the fluid routing plug shown in **FIG. 261**, taken along line CD-CD.

(288) **FIG. 268** is a cross-sectional view of the fluid routing plug shown in **FIG. 261**, taken along line CE-CE.

(289) **FIG. 269** is a cross-sectional view of the fluid routing plug shown in **FIG. 261**, taken along line CF-CF.

(290) **FIG. 270** is a cross-sectional view of the fluid routing plug shown in **FIG. 261**, taken along line CG-CG.

(291) **FIG. 271** is a side elevational and conical-section view of the second surface of the fluid routing plug shown in **FIG. 261**, the conical-section is taken from circle CI shown in **FIG. 262** to circle CJ shown in **FIG. 265**.

(292) **FIG. 272** is a top perspective view of the second surface of the fluid routing plug shown in **FIG. 271**.

(293) **FIG. 273** is a perspective view of the first surface of the fluid routing plug shown in **FIG. 271**.

(294) **FIG. 274** is a perspective view of the second surface of the fluid routing plug shown in **FIG. 271**.

(295) **FIG. 275** is a top plan view of another embodiment of a fluid routing plug.

(296) **FIG. 276** is a perspective view of the second surface of the fluid routing plug shown in **FIG.**

- 275.
- (297) FIG. 277 is a side elevational view of the second surface of the fluid routing plug shown in FIG. 275.
- (298) FIG. 278 is a cross-sectional view of the fluid routing plug shown in FIG. 277, taken along line CP-CP.
- (299) FIG. 279 is a perspective view of the first surface of the fluid routing plug shown in FIG. 275.
- (300) FIG. 280 is a side elevational view of the first surface of the fluid routing plug shown in FIG. 275.
- (301) FIG. 281 is a cross-sectional view of the fluid routing plug shown in FIG. 275, taken along line CK-CK.
- (302) FIG. 282 is a cross-sectional view of the fluid routing plug shown in FIG. 275, taken along line CM-CM.
- (303) FIG. 283 is a cross-sectional view of the fluid routing plug shown in FIG. 275, taken along line CN-CN.
- (304) FIG. 284 is the side elevational view of the first surface of the fluid routing plug shown in FIG. 280, but one of the suction fluid passages and one of the discharge fluid slots are shown in phantom.
- (305) FIG. 285 is a cross-sectional view of the fluid routing plug shown in FIG. 275, taken along line CL-CL. A portion of each of the suction fluid passages is shown in phantom.
- (306) FIG. 286 is a top perspective view of the fluid routing plug shown in FIG. 275.
- (307) FIG. 287 is an enlarged view of area CR shown in FIG. 286.
- (308) FIG. 288 is a side elevational and conical-section view of the second surface of the fluid routing plug shown in FIG. 275, the conical-section is taken from circle CO shown in FIG. 276 to circle CQ shown in FIG. 279.
- (309) FIG. 289 is top perspective view of the second surface of the fluid routing plug shown in FIG. 288.
- (310) FIG. 290 is a perspective view of the first surface of the fluid routing plug shown in FIG. 288.
- (311) FIG. 291 is a side elevational view of the fluid routing plug shown in FIG. 288.
- (312) FIG. 292 is a perspective view of the second surface of another embodiment of a fluid routing plug.

DETAILED DESCRIPTION

- (313) Turning now to the non-prior art figures, FIGS. 6 and 7 show a fluid end 100. The fluid end 100 may be attached to the traditional power end 34, shown in FIG. 3. Alternatively, the fluid end 100 may be attached to various embodiments of power ends, such as the modular power end described in U.S. Provisional Patent Application Ser. No. 63/053,797, authored by Thomas et al. and filed on Jul. 20, 2020.
- (314) Unlike the traditional fluid end 46, shown in FIG. 3, the fluid end 100 comprises a plurality of fluid end sections 102 rather than a single housing 48. The fluid end sections 102 are positioned in a side-by-side relationship. Preferably, the fluid end 100 comprises five fluid end sections 102. However, more or less fluid end sections 102 may be used. Forming the fluid end 100 out of multiple fluid end sections 102 allows a single fluid end section 102 to be replaced, if needed. In contrast, the entire housing 48 in traditional fluid ends 46 may need to be replaced if only a portion of the housing 48 fails.
- (315) Turning to FIGS. 8 and 9, each fluid end section 102 comprises a horizontally positioned housing 104 having a generally cylindrical cross-sectional shape, as shown in FIG. 8. In alternative embodiments, each fluid end section may have a generally rectangular cross-sectional shape. Unlike the traditional fluid end 46 shown in FIGS. 3 and 5, each housing 104 does not include a vertical bore intersecting a horizontal bore to form an internal chamber. Rather, each housing 104

only has a single horizontally positioned bore **106**, as shown in FIG. **9**. Removing the internal chamber found in traditional fluid ends from the housing **104** removes common stress points from the housing **104**.

(316) Eliminating the intersecting bore also reduces the cost of manufacturing the fluid end **100** as compared to traditional fluid ends. The time required to manufacture the fluid end **100** is greatly reduced without the need for machining an intersecting bore, and the fluid end **100** may be manufactured on a lathe instead of a machining center. The fluid end **100** may also be manufactured out of lower strength and less costly materials since it does not include the high stress areas found in traditional fluid ends. Each housing **104** may be manufactured out of high strength alloy steel, such as carbon steel. In alternative embodiments, each housing **104** may be manufactured out of stainless steel.

(317) Continuing with FIGS. **8** and **9**, each housing **104** comprises a first outer surface **108** joined to an opposed second outer surface **110** by an intermediate outer surface **112**. The horizontal bore **106** extends through the housing **104** along a central longitudinal axis **114** and interconnects the opposed first and second outer surfaces **108** and **110**, as shown in FIG. **9**. Each housing **104** is of single piece construction.

(318) Since each housing **104** only has a single horizontal bore **106**, fluid must be routed throughout the housing **104** differently from how fluid is routed throughout a traditional fluid end housing **48**. As will be described in more detail herein, a fluid routing plug **116**, shown in FIGS. **52-64**, is installed within each housing **104** and is configured to route fluid throughout the housing **104**.

(319) With reference to FIGS. **6**, **7**, and **10-16**, each housing **104** is supported on a single connect plate **118** in a one-to-one relationship. A plurality of sets of stay rods **120**, shown in FIG. **6**, are used to attach each connect plate **118** to a power end. The connect plates **118** may each be attached to the corresponding stay rods **120** prior to attaching a housing **104** to a corresponding connect plate **118**. Because the housings **104** are each attached to a connect plate **118**, the fluid end **100** does not include a flange like the flange **50** formed in the fluid end **46** shown in FIG. **3**. In an alternative embodiment, multiple housings may be attached to a single, larger connect plate. In such embodiment, the stay rods are likewise attached to the single, larger connect plate.

(320) The stay rods **120** shown in FIG. **6** are configured for use with a modular power end, like that shown in U.S. Provisional Patent Application Ser. No. 63/053,797, authored by Thomas et al. and filed on Jul. 20, 2020. A spacer **122** is installed on each stay rod **120** and is configured to engage with a front surface of the power end. In alternative embodiments, the stay rods may be configured like the stay rods **42** shown in FIG. **3**.

(321) With reference to FIGS. **10-13**, each connect plate **118** has a generally rectangular shape and has opposed first and second surfaces **124** and **126**. A plurality of first passages **128** are formed around the outer periphery of each connect plate **118**. Each first passage **128** interconnects the first and second surfaces **124** and **126** of the connect plate **118** and is configured for receiving a stay rod **120**. Each stay rod **120** extends through a corresponding passage **128** in a one-to-one relationship.

(322) The connect plate **118** shown in FIGS. **10-13** has four first passages **128**. Likewise, four stay rods **120** are shown attached to each connect plate **118** in FIG. **6**. In alternative embodiments, the connect plate may have more than four or less than four first passages, as long as the amount of first passages corresponds with the number of stay rods being used with each connect plate.

(323) Once each stay rod **120** is installed in a connect plate **118**, a first end **130** of each stay rod **120** projects from the first surface **124** of the connect plate **118**, as shown in FIG. **16**. A nut **132** and a washer **134** are installed on the projecting first end **130** of each stay rod **120** in a one-to-one relationship. The nut **132** is turned until it tightly engages a corresponding washer **134** and the first surface **124** of the connect plate **118**, thereby securing the connect plate **118** to the stay rods **120**.

(324) With reference to FIGS. **6**, and **14-16**, a plurality of notches **136** are formed around the periphery of the housing **104** at its second surface **110**, as shown in FIGS. **14** and **15**. When the

housing **104** is attached to the connect plate **118**, each notch **136** partially surrounds one of the first passages **128** in a one-to-one relationship. The notches **136** provide space to access the washer **134** and nut **132** during operation.

(325) Continuing with FIGS. **10-13**, a central bore **138** is formed in each connect plate **118** and interconnects the first and second surfaces **124** and **126**. The central bore **138** is configured for receiving a stuffing box **140**, as described in more detail later herein. A plurality of second passages **142** are formed in the connect plate **118** and surround the central bore **138**. Each second passage **142** interconnects the first and second surfaces **124** and **126** of the connect plate **118**. The second passages **142** are configured to align in a one-to-one relationship with a plurality of first threaded openings **144** formed in the second surface **110** of each housing **104**, as shown in FIGS. **14** and **15**.

(326) Each housing **104** is attached to the first surface **124** of a corresponding connect plate **118** using a fastening system **146**. The fastening system **146** comprises a plurality of studs **148**, a plurality of washers **150**, and a plurality of nuts **152**, as shown in FIGS. **7** and **17**. A first end **154** of each stud **148** is configured to mate with a corresponding one of the first openings **144** formed in the housing **104**. The second passages **142** formed in the connect plate **118** subsequently receive the plural studs **148** projecting from the housing **104**.

(327) When the housing **104** and the connect plate **118** are brought together, a second end **156** of each stud **148** projects from the second surface **126** of the connect plate **118**. A washer **150** and a nut **152** are subsequently installed on the second end **156** of each stud **148**, in a one-to-one relationship. The nut **152** is turned until it tightly engages the washer **150** and the second surface **126** of the connect plate **118**, thereby securing the housing **104** and the connect plate **118** together.

(328) In FIGS. **10-15**, the housing **104** and connect plate **118** each have eight corresponding first openings **144** and second passages **142**. In alternative embodiments, more than eight or less than eight corresponding openings and second passages may be formed in the housing and connect plate. In such embodiments, the fastening system may comprise the same number of studs, washers, and nuts as there are openings and passages. In further alternative embodiments, the fastening system may comprise different types of fasteners, such as socket-headed screws.

(329) Continuing with FIGS. **10-15**, a pair of third passages **158** are formed in the connect plate **118** on opposite sides of the central bore **138**. The third passages **158** are alignable with a pair of pin holes **160** formed in the second surface **110** of the housing **104**. Each third passage **158** and each corresponding pin hole **160** is configured to receive a dowel pin in a one-to-one relationship. The dowel pins are used to help align the housing **104** on the connect plate **118** during assembly. A threaded hole **162** may also be formed in a top surface **164** of each connect plate **118**, as shown in FIGS. **10** and **11**. The threaded hole **162** is configured for receiving a lifting eye (not shown) used to lift and support the connect plate **118** during assembly.

(330) In alternative embodiments, the connect plate may have various shapes and sizes other than those shown in FIGS. **10-13**. For example, the connect plate may be shaped like the various embodiments disclosed in U.S. Provisional Patent Ser. No. 63/053,797, authored by Thomas et al. and filed on Jul. 20, 2020.

(331) Turning back to FIGS. **6** and **7**, in contrast to the traditional fluid end **46**, shown in FIG. **3**, the fluid end **100** is configured to receive fluid from two manifolds, rather than just one. The fluid end **100** comprises an upper intake manifold **166** and a lower intake manifold **168**. Each manifold **166** and **168** is in fluid communication with each fluid end section **102**. Using two different manifolds **166** and **168** allows different types of fluid to be delivered to each fluid end section **102**. For example, fluid having a higher level of proppant may be delivered via the upper intake manifold **166**, while fluid having a zero to minimal level of proppant may be delivered via the lower intake manifold **168**.

(332) Continuing with FIGS. **6** and **7**, the upper and lower intake manifolds **166** and **168** are joined to the fluid end sections **102** via a plurality of conduits **159**. Each conduit **159** is positioned directly below the corresponding manifold **166** and **168** and extends along a straight line between the fluid

end section **102** and the corresponding manifold **166** and **168**. Thus, each conduit **159** and corresponding manifolds **166** and **168** have a “T” shape.

(333) Turning to FIGS. 7A and 7B, an alternative embodiment of an upper and lower intake manifold **161** and **163** is shown. The upper and lower intake manifolds **161** and **163** are joined to the fluid end sections **102** via a plurality of conduits **165**. The conduits **165** have an elbow shape. The elbow shape of the conduits **165** causes the corresponding manifolds **161** and **163** to be spaced farther away from a discharge manifold **167**, than the manifolds **166** and **168**. Providing more space between the intake manifolds **161** and **163** and the discharge manifold **167** provides more space for maintenance to different areas of the fluid end **100**, when needed.

(334) Turning back to FIG. 9, an upper and lower intake bore **170** and **172** are formed within the housing **104**. Each bore **170** and **172** interconnects the intermediate outer surface **112** and the horizontal bore **106**. The upper and lower intake bores **170** and **172** shown in FIG. 9 are collinear. In alternative embodiments, the upper and lower intake bores may not be collinear.

(335) With reference to FIGS. 6-9, the upper intake bore **170** is in fluid communication with the upper intake manifold **166**, and the lower intake bore **172** is in fluid communication with the lower intake manifold **168**. In operation, fluid may be delivered into the housing **104** through both the upper and lower intake bores **170** and **172**. In alternative embodiments, only one intake bore may be formed in the housing and only one intake manifold may be attached to the housing.

(336) Continuing with FIGS. 6-9, the fluid end **100** further comprises a plurality of discharge conduits **174**. Each discharge conduit **174** is attached to one of the fluid end sections **102** in a one-to-one relationship. A discharge manifold **176** interconnects each of the discharge conduits **174**, as shown in FIGS. 6 and 7. In alternative embodiments, the discharge conduits and discharge manifold may be formed as a single unit, like the discharge manifold **167**, shown in FIGS. 7A and 7B.

(337) Continuing with FIG. 9, a discharge bore **178** is formed in the housing **104** and interconnects the intermediate surface **112** and the horizontal bore **106**. The discharge bore **178** is positioned between the first surface **108** of the housing **104** and the intake bores **170** and **172**. The discharge bore **178** is in fluid communication with the discharge conduit **174**. In operation, fluid to be pressurized enters the housing **104** through the upper and lower intake bores **170** and **172**. Pressurized fluid exits the housing **104** through the discharge bore **178**.

(338) With reference to FIG. 18, the discharge bore **178** has an oval cross-sectional shape, as shown by a discharge bore opening **180**. The opening **180** has a length A and a width B. The discharge bore **178** is formed within the housing **104** such that the width B extends along an axis that is parallel to the longitudinal axis **114** of the housing **104**. During operation, high fluid pressure within the discharge bore **178** may cause the walls along the length A to compress, causing the discharge bore **178** to have a more circular cross-sectional shape. Providing room for the walls surrounding the discharge bore **178** to compress, helps reduce stress in the housing **104** and increase fluid flow. In alternative embodiments, the discharge bore may have a circular cross-sectional shape.

(339) Continuing with FIG. 19, a counterbore **173** is formed within the housing **104** immediately above the opening **180** of the discharge bore **178**. The discharge bore **178** opens into the counterbore **173**. The counterbore **173** has a circular cross-sectional shape, as shown by the opening **175** in FIG. 18. A portion of the discharge conduit **174** is installed within the counterbore **173** through its opening **175**. A seal **182** is interposed between the walls of the housing **104** surrounding the discharge bore **178** and an outer surface of the discharge conduit **174**. The seal **182** is installed within a groove **184** formed in the walls of the housing **104**. The seal **182** may be identical to the second seal **376**, described with reference to FIGS. 65 and 70. In alternative embodiments, the seal may be identical to the first seal **374**, described with reference to FIGS. 65 and 71.

(340) The groove **184** is characterized by two sidewalls **185** joined to a base **183**. The sidewalls

185 may join the base **183** via radius corners or at a 90 degree angle. No grooves are formed in the outer surface of the discharge conduit **174** for housing a seal. In operation, the seal **182** wears against the outer surface of the discharge conduit **174**. If the outer surface of the discharge conduit **174** begins to erode, allowing fluid to leak around the seal **182**, the discharge conduit **174** may be replaced with a new discharge conduit **174**.

(341) The discharge bore **178** shown in FIG. **9** interconnects a top surface **113** of the intermediate surface **112** of the housing **104** and the horizontal bore **106**. Likewise, the discharge conduits **174** shown in FIGS. **6**, **7**, and **9** are attached to the top surface **113** of the intermediate surface **112** of each housing **104**. In operation, any gas trapped within the housing **104** rises towards the top of the housing **104**. Placing the discharge bore **178** and conduit **174** at the top of the housing **104** allows the gases to naturally escape. Additionally, any wear caused to the components by the rising gas will primarily be imposed on the discharge conduit **174**, rather than the housing **104**. The discharge conduit **174** and corresponding discharge piping **176** are easily replaced, if needed.

(342) In alternative embodiments, the discharge bore may interconnect a bottom or side surface of the intermediate surface and the horizontal bore, and the discharge conduit may be attached to the corresponding surface of the housing. In further alternative embodiments, the discharge bore may interconnect the first outer surface of the housing and the horizontal bore, and the discharge conduit may be attached to the first outer surface of the housing.

(343) With reference to FIGS. **6**, **18** and **19**, a rectangular flange **171** is formed around each discharge conduit **174**. Each rectangular flange **171** is attached to the housing **104** using a plurality of threaded studs **186** and nuts **187**, as shown in FIGS. **6** and **19**. A plurality of threaded openings **188** are formed in the housing **104** for receiving the studs **186**, as shown in FIG. **18**. The openings **188** are positioned in a rectangular pattern around the discharge bore opening **180**. Such pattern helps maximize the surface area of the intermediate surface **112** of the housing **104**, helping to reduce the size and weight of the housing **104**.

(344) With reference to FIGS. **7** and **18**, the intake manifolds **166** and **168** each comprise a plurality of rectangular flanges **189** joined to a plurality of conduits **191** in a one-to-one relationship, as shown in FIG. **7**. Each rectangular flange **189** is attached to the housing **104** using a plurality of threaded studs **190** and nuts **193**, as shown in FIG. **7**. A plurality of threaded openings **192** are formed in the housing **104** for receiving the studs **190**, as shown in FIG. **18**. The openings **192** are positioned in a rectangular pattern around the intake bores **170** and **172** to maximize surface area of the housing **104**. In alternative embodiments, the discharge conduits and intake manifolds may be attached to the housing using different types of fasteners, such as socket-headed screws.

(345) Continuing with FIG. **18**, the intermediate surface **112** of the housing **104** includes a first portion **194** joined to a second portion **196** by a first tapered portion **198**. The second portion **196** is joined to a third portion **200** by a second tapered portion **202**. The first portion **194** is joined to the first surface **108** and the third portion **200** is joined to the second surface **110**.

(346) The second portion **196** has a smaller diameter than both the first and third portions **194** and **200**. Providing the second portion **196** with a smaller diameter helps remove unnecessary weight from the housing **104**. The third portion **200** may have a slightly larger diameter than the first portion **194**. The first, second, and third portions **194**, **196**, and **200** are generally cylindrical. Thus, the housing **104** may be characterized as being primarily cylindrical. In alternative embodiments, the housing may be uniform in diameter throughout its intermediate surface. In further alternative embodiments, the housing may have various diameters throughout its intermediate surface other than those shown in FIG. **18**.

(347) Continuing with FIG. **18**, a threaded hole **204** is formed in the top surface **113** of the intermediate surface **112**. The threaded hole **204** is positioned at the center of gravity of the housing **104** when the housing **104** is fully loaded with the components described herein. The threaded hole **204** is configured to receive a lifting eye (not shown) used to lift and support the housing **104** during assembly and maintenance, as shown in FIG. **9**.

(348) With reference to FIGS. 20-29, each fluid end section 102 further comprises a stuffing box 140 attached to the second outer surface 110 of the housing 104. The stuffing box 140 has a generally cylindrical shape and comprises a first outer surface 206 joined to an opposed second outer surface 208 by an intermediate outer surface 210. The intermediate surface 210 includes a cylindrical first portion 212 joined directly to a cylindrical second portion 214. The first portion 212 is positioned adjacent the first surface 206 and has a reduced diameter from that of the second portion 214. A threaded hole 215 is formed in a top surface of the second portion 214. The threaded hole 215 is configured to receive a lifting eye (not shown) used to lift and support the stuffing box 140 during assembly and maintenance.

(349) A central passage 216 interconnects the stuffing box's first and second outer surfaces 206 and 208. The walls surrounding the central passage 216 include a first section 218 joined to a second section 220 by a tapered shoulder 222, as shown in FIGS. 25, 26, and 29. The second section 220 has a larger diameter than that of the first section 218. As described in more detail herein, the second section 220 and the tapered shoulder 222 are configured for receiving a plunger packing 224, as shown in FIGS. 20 and 21.

(350) Continuing with FIGS. 23-29, a plurality of passages 226 are formed around the periphery of the second portion 214 of the stuffing box 140. Each passage 226 interconnects the second surface 208 of the stuffing box 140 and a base 228 of the second portion 214. The passages 226 are formed parallel to the central passage 216.

(351) Turning back to FIGS. 14 and 15, a plurality of second threaded openings 230 are formed in the second surface 110 of the housing 104. The openings 230 surround the opening of the horizontal bore 106. The second openings 230 are surrounded by the first openings 144 used with the connect plate 118.

(352) Continuing with FIGS. 20 and 21, the walls surrounding the horizontal bore 106 adjacent the second surface 110 of the housing 104 are sized to receive the first portion 212 of the stuffing box 140. The first portion 212 is installed within the horizontal bore 106 such that the base 228 of the second portion 214 abuts the second surface 110 of the housing 104. A portion of the second portion 214 is disposed within the central bore 138 formed in the connect plate 118. The stuffing box 140 is aligned on the housing 104 such that the passages 226 align with the second openings 230 in a one-to-one relationship.

(353) With reference to FIGS. 20, 21, and 30-37, the stuffing box 140 is attached to the housing 104 using a retainer 232 and a fastening system 234. The retainer 232 has a generally cylindrical shape and comprises opposed first and second outer surfaces 236 and 238 joined by an intermediate surface 240. A central passage 242 interconnects the first and second outer surfaces 236 and 238. At least a portion of the central passage 242 has internal threads 244. A plurality of side passages 246 are formed in the retainer 232. Each passage 246 interconnects the central passage 242 and the intermediate surface 240. The passages 246 provide a pathway for lubricating oil to be introduced to the horizontal bore 106 during operation. The oil lubricates the moving parts within the housing 104 during operation.

(354) Continuing with FIGS. 30-37, a plurality of passages 248 are formed in the retainer 232 and surround the central passage 242. Each passage 248 interconnects the first and second outer surfaces 236 and 238. The first surface 236 of the retainer 232 is positioned on the second surface 208 of the stuffing box 140 such that the passages 248 align with the passages 226 formed in the stuffing box 140, in a one-to-one relationship.

(355) A pair of dowel pin holes 241 are formed in the second surface 208 of the stuffing box 140, as shown in FIGS. 27 and 28. A corresponding pair of dowel pin holes 243 are formed in the first surface 236 of the retainer 232, as shown in FIGS. 31 and 32. The holes 241 and 243 are configured for receiving a dowel pin. The dowel pin aligns the retainer 232 on the stuffing box 140 during assembly.

(356) Turning back to FIGS. 20 and 21, the fastening system 234 secures both the retainer 232 and

the stuffing box **140** to the housing **104**. The fastening system **234** comprises a plurality of studs **250**, nuts **252**, and washers **254**. A first end **256** of each stud **250** mates with one of the second openings **230** in the housing **104** in a one-to-one relationship. The passages **226** in the stuffing box **140** and the passages **248** in the retainer **232** subsequently receive the plural studs **250** projecting from the housing **104**.

(357) A second end **258** of each stud **250** projects from the second surface **238** of the retainer **232**. The projecting second end **258** of each stud **250** receives a washer **254** and a nut **252**. The nut **252** is turned until it tightly engages the washer **254** and the second surface **238** of the retainer **232**, thereby securing the retainer **232** and the stuffing box **140** together. The retainer **232**, in turn, holds the stuffing box **140** against the housing **104**. The stuffing box **140** and the retainer **232** may be attached to and removed from the housing **104** without removing the connect plate **118**.

(358) When the first portion **212** of the stuffing box **140** is installed within the housing **104**, a seal **260** is interposed between the walls of the housing **104** and outer surface of the first portion **212**. The seal **260** is installed within a groove **262** formed in the walls of the housing **104**. The seal **260** may be identical to the first seal **374**, described with reference to FIGS. **65** and **71**. In alternative embodiments, the seal may be identical to the second seal **376**, described with reference to FIGS. **65** and **70**.

(359) The groove **262** is characterized by two sidewalls **264** joined by a base **266**, as shown in FIG. **21**. The sidewalls **264** may join the base **266** via radius corners or at a 90 degree angle. No grooves are formed in the first portion **212** of the stuffing box **140** for housing a seal. The seal **260** wears against the outer surface of the first portion **212** during operation. If the outer surface of the first portion **212** begins to erode, allowing fluid to leak around the seal **260**, the stuffing box **140** may be replaced with a new stuffing box **140**.

(360) When the stuffing box **140** is attached to the housing **104** using the fastening system **234**, a first end **256** of the studs **250** may be installed within the housing **104** such that they extend past the seal **260**, as shown in FIG. **20**. An edge of the studs **250** may not be purposely aligned with an edge of the seal **260** in order to prevent areas of high stress from being aligned with one another in the housing **104**, potentially causing a stress riser.

(361) Continuing with FIGS. **20**, **21**, and **38-42**, a plunger packing **224** is installed within the central passage **216** of the stuffing box **140**. The plunger packing **224** engages the tapered shoulder **222** and is positioned within the second section **220** of the central passage **216**, as shown in FIGS. **20** and **21**. A portion of the plunger packing **224** may extend into the central passage **242** of the retainer **232**. The plunger packing **224** has a central passage **268** that aligns with the central passages **216** and **242** when the plunger packing **224** is installed within the stuffing box **140** and the retainer **232**. In alternative embodiments, the plunger packing may be sized to not extend into the retainer.

(362) The plunger packing **224** comprises a pair of outer ring seals **270** and **271** and at least one inner ring seal **272**. The outer ring seals **270** and **271** may be made of metal while the inner ring seals **272** may be made of an elastomer material. The outer ring **270** has a tapered outer surface **274** that is sized to engage the tapered shoulder **222** formed in the central passage **216**. The tapered engagement helps reduce stress in the stuffing box **140** during operation. In alternative embodiments, the walls surrounding the central passage of the stuffing box may include an annular shoulder rather than a tapered shoulder. In such embodiment, the plunger packing may have a flat outer ring configured to mate with the annular shoulder. A plurality of holes **275** are formed in the outer ring **271**. The holes **275** are in fluid communication with the side passages **246** formed in the retainer **232** in order to deliver lubricating oil to the housing **104**.

(363) With reference to FIGS. **20**, **21**, and **43-46**, a packing nut **276** is installed within the retainer **232** and engages the plunger packing **224**. The packing nut **276** comprises a first surface **278** joined to an opposed second surface **280** by an intermediate surface **282**. A central passage **284** extends through the packing nut **276** and interconnects the opposed first and second surfaces **278** and **280**.

A plurality of side holes **286** are formed in the packing nut **276** and interconnect the central passage **284** and the intermediate surface **282**. The holes **286** are configured for engaging a tool used to grip the packing nut **276**.

(364) Continuing with FIGS. **43-46**, external threads **288** are formed in a portion of the intermediate surface **282** of the packing nut **276**. The external threads **288** are configured to mate with the internal threads **244** formed within the retainer **232**, as shown in FIGS. **20** and **21**. The mating threads **288** and **244** are buttress threads. The buttress threads are configured to handle a large amount of load using a low number of threads. Using a low number of threads allows the packing nut **276** to be quickly removed or installed within the retainer **232**. In alternative embodiments, the packing nut and retainer may mate using traditional threads.

(365) When the packing nut **276** is installed within the retainers **232**, the first surface **278** of the packing nut **276** engages an outer ring seal **270** of the plunger packing **224**. Such engagement compresses the plunger packing **224**, creating a tight seal. After the packing nut **276** has been installed within a retainer **232**, the central passage **284** within the packing nut **276** is aligned with the central passage **268** in the plunger packing **224**.

(366) Continuing with FIGS. **20** and **21**, when the stuffing box **140** and the retainer **232** are attached to the housing **104**, the central passages **216** and **242** align with the horizontal bore **106**. Likewise, the central passages **268** and **284** in the installed plunger packing **224** and packing nut **276** align with the horizontal bore **106**. Thus, the central passages **216**, **242**, **268**, and **284** may be considered an extension of the horizontal bore **106**. A plunger **290** is disposed with the installed plunger packing **224** and the packing nut **276**, as shown in FIG. **20**. In operation, the plunger **290** reciprocates within the horizontal bore **106** in order to pressurize fluid contained within the housing **104**.

(367) With reference to FIGS. **20**, **21**, and **47-49**, the horizontal bore **106** is sealed at the first surface **108** of the housing **104** by a retainer **300**. The retainer **300** has a first surface **302** joined to an opposed second surface **304** by an outer intermediate surface **306**. A cutout **308** is formed in the second surface **304** for receiving a portion of a discharge valve guide **298**. A central passage **310** is formed in the retainer **300** and interconnects the first surface **302** and the cutout **308**. The walls surrounding the central passage **310** have a polygonal shape. The polygonal shape is configured to mate with a tool used to grip the retainer **300**.

(368) The intermediate surface **306** of the retainer **300** has external threads **312** that mate within internal threads **314** formed in the walls surrounding the horizontal bore **106** adjacent the first surface **108** of the housing **104**, as shown in FIGS. **20** and **21**. The mating threads **312** and **314** are buttress threads. The buttress threads are configured to handle a large amount of load using a low number of threads. Using a low number of threads allows the retainer **300** to be quickly removed from or installed within the housing **104**. In alternative embodiments, the retainer may mate with the housing using traditional threads. In further alternative embodiments, the retainer may be secured to the housing using a fastening system, like the fastening system **234**.

(369) Turning now to FIGS. **50** and **51**, the fluid routing plug **116** is installed within a medial section of the horizontal bore **106**. The fluid routing plug **116** is configured to engage with a suction valve **292** on one side and a discharge valve **294** on the opposite side. In operation, the suction and discharge valves **292** and **294** move axially along an axis that is parallel to or aligned within the central longitudinal axis **114** of the housing **104**, shown in FIG. **9**, as the valves **292** and **294** move at alternating times between an open and closed position. In the closed position, the valves **292** and **294** are pressed against the fluid routing plug **116**, preventing fluid from exiting the plug **116**. In the open position, the valves **292** and **294** are spaced from the fluid routing plug **116**, allowing fluid to flow from the plug **116**.

(370) As will be described in more detail herein, axial movement of the suction valve **292** is limited by a suction valve guide **296** installed within the housing **104**. Likewise, axial movement of the discharge valve **294** is limited by the discharge valve guide **298** installed within the housing **104**.

(371) Turning now to FIGS. **52-64**, the fluid routing plug **116** comprises a body **316** having opposed first and second outer surfaces **318** and **320** joined by an intermediate outer surface **322**. The first outer surface **318** may also be referred to as the suction side of the fluid routing plug **116**. The second outer surface **320** may also be referred to as the discharge side of the fluid routing plug **116**. A central longitudinal axis **324** extends through the body **316** and both surfaces **318** and **320**, as shown in FIG. **55**.

(372) A plurality of first fluid passages **326** are formed within the body **316** and interconnect the intermediate surface **322** and the first surface **318**. The first fluid passages **326** interconnect the intermediate surface **322** and the first surface **318** by way of an axial-blind bore **328**, as shown in FIG. **55**. The blind bore **328** extends along the central longitudinal axis **324** of the body **316**. The first fluid passages **326** each open into the blind bore **328** via a plurality of openings **330**. A longitudinal axis **332** of each first fluid passage **326** intersects the central longitudinal axis **324** of the body **316**, as shown in FIG. **58**.

(373) The fluid routing plug **116** shown in FIGS. **52-64** has four first fluid passages **326** formed in its body **316**. The first fluid passages **326** are equally spaced around the body **316**. In alternative embodiments, more than four or less than four first fluid passages may be formed in the body and may be equally or unequally spaced apart from one another.

(374) Continuing with FIG. **55**, the first fluid passages **326** extend between the intermediate surface **322** and the blind bore **328** at a non-right angle relative to the central longitudinal axis **324**—the acute angle facing the second surface **320** of the body **316**. Forming the first fluid passages **326** at such an angle reduces the amount of stress in the fluid routing plug **116** as fluid flows through the first fluid passages **326**. Forming the first fluid passages **326** at such angle also helps direct fluid flow towards the blind bore **328** and the first surface **318**.

(375) With reference to FIGS. **57** and **59**, the first fluid passages **326** have an oval cross-sectional shape, as shown by an opening **334** of each first fluid passage **326** on the intermediate surface **322**. Each opening **334** has a length A and a width B, as shown in FIG. **59**. The first fluid passages **326** are formed in the body **316** such that the length A extends along an axis that is parallel to the central longitudinal axis **324** of the body **316**. Orienting the first fluid passages **326** as such helps reduce the amount of stress in the body **316** as fluid flows through the first fluid passages **326** and helps maximize the rate of fluid flow through the passages **326**. In alternative embodiments, the first fluid passages may have a different cross-sectional shape, such as a circular or oblong shape. In further alternative embodiments, the first fluid passages may be shaped like the first fluid passages **910**, shown in FIGS. **121** and **124**.

(376) With reference to FIGS. **60-63**, the fluid routing plug **116** further comprises a plurality of second fluid passages **336** formed in the body **316**. The second fluid passages **336** each have a circular cross-sectional shape and interconnect the first and second surfaces **318** and **320** of the body **316**. In alternative embodiments, the second fluid passages may have a different cross-sectional shape, such as an oval or oblong shape.

(377) Unlike the first fluid passages **326**, the second fluid passages **336** do not intersect an axially blind bore. Rather, each second fluid passage **336** extends between the first and second surface **318** and **320** along a straight-line path. The second fluid passages **336** and the first fluid passages **326** do not intersect and are positioned offset from one another, as shown in FIG. **58**. Positioning the first and second passages **326** and **336** offset from one another helps minimize the stress in the fluid routing plug **116** during operation. The fluid routing plug **116** shown in FIGS. **52-64** has twelve second fluid passages **336** formed in its body **316**. In alternative embodiments, more or less than twelve second fluid passages may be formed in the body.

(378) Each second fluid passage **336** extends between the first and second surfaces **318** and **320** along a different axis, as shown in FIGS. **60-63**. Each axis is positioned at a non-zero angle relative to the central longitudinal axis **324** of the body **316**. Forming each second passage **336** along a different axis helps alleviate stress in the fluid routing plug **116** during operation and helps

maximize the rate of fluid flow through the second passages 336.

(379) Turning back to FIGS. 53, 55, and 56, the first surface 318 of the body 316 includes an outer rim 338 joined to a tapered wall 340. The outer rim 338 may taper slightly between the intermediate surface 322 and the tapered wall 340, as shown in FIG. 55. Such taper provides more surface area for the tapered wall 340 without increasing the length of the intermediate surface 322. The tapered wall 340 extends between an entrance 342 of the blind bore 328 and the outer rim 338 at an angle of at least 30 degrees relative to the central longitudinal axis 324 of the body 316.

Preferably, the tapered wall 340 is formed at an angle of 45 degrees relative to the central longitudinal axis 324 of the body 316, as is shown in FIG. 55. As will be described in more detail later herein, the tapered wall 340 forms a cavity 344 within the first surface 318 of the body 316 that is sized to receive a sealing element 346 of the suction valve 292, as shown in FIGS. 72-76.

(380) Continuing with FIGS. 53 and 56, the second fluid passages 336 open on the outer rim 338 of the first surface 318, as shown by the openings 348. The second fluid passages 336 are formed within the body 316 such that the openings 348 are positioned in groups 350 around the outer rim 338. The first surface 318 shown in FIG. 59 comprises four groups 350 of openings 348, each group 350 comprising three openings 348. Adjacent openings 348 within each group 350 are equally spaced. The spacing between the nearest openings 348 of adjacent groups 350 exceeds the spacing between adjacent openings 348 within a single group 350. Spacing the openings 348 in groups 350 helps achieve the ideal velocity of fluid flow through the fluid routing plug 116 and allows the second fluid passages 336 to be offset from the first fluid passages 326, as shown in FIG. 58. In alternative embodiments, the openings may be spaced in differently sized groups or different patterns than that shown in FIG. 56.

(381) With reference to FIGS. 52, 54, and 55, the second surface 320 of the body 316 comprises an outer rim 352 joined to a central base 354 by a tapered wall 356. The tapered wall 356 extends between the central base 354 and the outer rim 352 at an angle of at least 30 degrees relative to the central longitudinal axis 324 of the body 316. Preferably, the tapered wall 356 is formed at an angle of 45 degrees relative to the central longitudinal axis 324 of the body 316, as is shown in FIG. 55. As will be described in more detail later herein, the tapered wall 356 forms a cavity 358 within the second surface 320 of the body 316 that is sized to receive a sealing element 360 of the discharge valve 294, as shown in FIGS. 85-89.

(382) Continuing with FIGS. 52, 54, and 55, a blind bore 362 is formed in the center of the central base 354. The walls surrounding the blind bore 362 may be configured to mate with a tool used to grip the fluid routing plug 116. For example, the walls surrounding the blind bore 362 may be threaded. The second fluid passages 336 open on the central base 354 of the second surface 320, as shown by the openings 364 in FIGS. 52 and 54. The second fluid passages 336 are formed within the body 316 such that the openings 364 surround the opening of the blind bore 362. The openings 364 shown in FIG. 54 are all equally spaced from one another around the opening of the blind bore 362. In alternative embodiments, the openings of the second fluid passages on the central base may not all be equally spaced apart from one another.

(383) Continuing with FIG. 55, in order to provide space for the openings 364 on the second surface 320, the tapered wall 356 has a greater diameter than the tapered wall 340 formed in the first surface 318. Thus, as will be described in more detail herein, the sealing element 360 of the discharge valve 294 is larger in size than the sealing element 346 of the suction valve 292, as shown in FIGS. 72-76 and 85-89.

(384) Turning back to FIGS. 50 and 51, the fluid routing plug 116 is installed within the horizontal bore 106 such that the first fluid passages 326 are in fluid communication with the upper and lower intake bores 170 and 172. The upper and lower intake bores 170 and 172 direct fluid into the first fluid passages 326 of the fluid routing plug 116. The first fluid passages 326 direct the fluid into the blind bore 328 and towards the first surface 318 of the fluid routing plug 116.

(385) When the plunger 290 is retracted from the housing 104, the fluid flowing through the first

fluid passages **326** forces the suction valve **292** to move axially away from the first surface **318**. Such position is considered an open position of the suction valve **292**. When the suction valve **292** is spaced from the first surface **318**, fluid flows out of the blind bore **328**, through the gap between the first surface **318** and the suction valve **292**. From there, the fluid flows around the suction valve **292** and the suction valve guide **296** and into the horizontal bore **106**. A first fluid flow path for the fluid to be pressurized is shown by the arrows **366** in FIG. **50**.

(386) With reference to FIG. **51**, as the plunger **290** extends into the horizontal bore **106**, the plunger **290** forces fluid in the horizontal bore **106** back towards the fluid routing plug **116**. Pressurized fluid forced back towards the fluid routing plug **116** by the plunger **290** forces the suction valve **292** to seal against the first surface **318**, sealing the entrance **342** of the blind bore **328**. Such position is considered a closed position of the suction valve **292**. Once the entrance **342** of the blind bore **328** is sealed, the only place for fluid to flow is through the openings **348** of the second fluid passages **336** on the outer rim **338** of the first surface **318**.

(387) Fluid flows into the openings **348** on the first surface **318** and through the second passages **336** towards the second surface **320** of the fluid routing plug **116**. The pressurized fluid at the second surface **320** forces the discharge valve **294** to move axially away from the second surface **320**, unsealing the openings **364** of the second fluid passages **336**. Such position is considered an open position of the discharge valve **294**. Pressurized fluid is then allowed to flow around the discharge valve **294** and into the discharge bore **178**. A second fluid flow path for the pressurized fluid is shown by the arrows **368** in FIG. **51**.

(388) When the plunger **290** retracts from the housing **104**, the fluid pressure on the back side of the discharge valve **294** is greater than the fluid pressure within the fluid routing plug **116**. Such pressure differential causes the discharge valve **294** to seal against the second surface **320**, sealing the openings **364** of the second fluid passages **336**. Such position is considered the closed position of the discharge valve **294**.

(389) Turning to FIG. **64**, the intermediate surface **322** of the fluid routing plug **116** varies in diameter throughout its length and generally decreases in size from its second surface **320** to its first surface **318**. The intermediate surface **322** comprises a first sealing surface **370** positioned adjacent the first surface **318** and a second sealing surface **372** positioned adjacent the second surface **320**. The first and second sealing surfaces **370** and **372** each extend around the entire intermediate surface **322** in an endless manner and surround the longitudinal axis **324** of the body **316**. The first and second sealing surfaces **370** and **372** shown in FIG. **64** are annular. In alternative embodiments, the first and second sealing surfaces may have non-annular shape, such as an oval shape.

(390) The first sealing surface **370** has a smaller diameter than the second sealing surface **372**. As will be described in more detail herein, the first and second sealing surfaces **370** and **372** are configured to engage a first and second seal **374** and **376** installed within the housing **104**, as shown in FIGS. **70** and **71**.

(391) Continuing with FIG. **64**, the intermediate surface **322** of the fluid routing plug **116** further comprises a first bevel **378** positioned between the opening **334** of the first fluid passages **326** and the first sealing surface **370**. The first bevel **378** extends around the entire intermediate surface **322** in an endless manner and surrounds the longitudinal axis **324** of the body **316**. The first bevel **378** shown in FIG. **64** is annular. In alternative embodiments, the first bevel may have a non-annular shape, such as an oval shape.

(392) A maximum diameter of the first bevel **378** is greater than the diameter of the first sealing surface **370**. The maximum diameter of the first bevel **378** is positioned adjacent the openings **334** of the first fluid passages **326** and a minimum diameter of the first bevel **378** is positioned adjacent the first sealing surface **370**. As will be described in more detail later herein, the first bevel **378** corresponds with a first beveled surface **380** formed in the housing **104**, as shown in FIGS. **65** and **69**.

(393) The intermediate surface **322** also comprises a second bevel **382** positioned between the second sealing surface **372** and the openings **334** of the first fluid passages **326**. The second bevel **382** extends around the entire intermediate surface **322** in an endless manner and surrounds the longitudinal axis **324** of the body **316**. The second bevel **382** shown in FIG. **64** is annular. In alternative embodiments, the first bevel may have non-annular shape, such as an oval shape.

(394) A maximum diameter of the second bevel **382** is positioned adjacent the second sealing surface **372** and a minimum diameter of the second bevel **382** is positioned adjacent the openings **334** of the first fluid passages **326**. The second sealing surface **372** and the maximum diameter of the second bevel **382** both have a greater diameter than the maximum diameter of the first bevel **378** and the diameter of the first sealing surface **370**.

(395) As will be described in more detail later herein, the second bevel **382** corresponds with a second beveled surface **384** formed in the housing **104**, as shown in FIGS. **65** and **68**. A small transition bevel **386** may extend between the second sealing surface **372** and the second bevel **382**. However, the transition bevel **386** does not engage the second beveled surface **384**, as shown in FIG. **68**. The transition bevel **386** helps reduce friction between the fluid routing plug **116** and the housing **104** during installation.

(396) As described above, the first and second bevels **378** and **382** are positioned between the first and second sealing surfaces **370** and **372**. The first and second bevels **378** and **382** help alleviate stress in the fluid routing plug **116** during operation. In alternative embodiments, the intermediate surface may only include a single bevel positioned between the first and second sealing surfaces.

(397) Continuing with FIG. **64**, the various diameters of the intermediate surface **322** are shown in more detail. The first sealing surface **370** has a diameter **D1**. The maximum diameter of the first bevel **378** has a diameter **D2**. The maximum diameter of the second bevel **382** has a diameter **D3**, and the second sealing surface **372** has a diameter **D4**. As described above in detail, **D4** is greater than **D3**, **D3** is greater than **D2**, and **D2** is greater than **D1**.

(398) With reference to FIG. **65**, in addition to being shaped to alleviate stress, the intermediate surface **322** is shaped to allow for easy installation of the fluid routing plug **116** within the horizontal bore **106**. The fluid routing plug **116** is installed into the horizontal bore **106** at the first outer surface **108** of the housing **104**. The fluid routing plug **116** is installed with the first surface **318** entering the horizontal bore **106** before the second surface **320**. The fluid routing plug **116** is pushed into the horizontal bore **106** until the first sealing surface **370** engages the first seal **374** and the second sealing surface **372** engages the second seal **376**.

(399) The first sealing surface **370** and first bevel **378** have smaller diameters than the second seal **376** and the second beveled surface **384**. Thus, clearance exists between these features as the fluid routing plug **116** is installed into the horizontal bore **106**. Providing such clearance during installation avoids unnecessary wear to both the housing **104** and fluid routing plug **116** during installation.

(400) With reference to FIGS. **65** and **67**, once the fluid routing plug **116** is installed within the housing **104**, an annular chamber **388** is formed between the walls of the housing **104** and the intermediate surface **322**. The intake bores **170** and **172** open into the chamber **388**. Only a couple of the openings **334** of the first fluid passages **326** may align with the intake bores **170** and **172**. Alternatively, the fluid routing plug **116** may be installed within the housing **104** such that none of the openings **334** directly align with the intake bores **170** and **172**. The chamber **388** provides a pathway for fluid from the intake bores **170** and **172** to flow around the fluid routing plug **116** and into the openings **334** of the first fluid passages **326**. The chamber **388** also provides space for proppant or other debris to collect during operation.

(401) Continuing with FIG. **67**, the walls of the housing **104** surrounding the horizontal bore **106** immediately adjacent the intake bores **170** and **172** are beveled, as shown by bevels **390** and **392**. The bevels **390** and **392** help reduce stress in the housing **104** during operation and increase the size of the annular chamber **388**. In alternative embodiments, the bevels **390** and **392** may be larger

than those shown in FIG. 67 in order to increase the size of the chamber 388, as shown for example in FIG. 100F. Similarly, the walls of the housing 104 surrounding the horizontal bore 106 immediately adjacent the discharge bore 178 are also beveled, as shown by the bevel 394 in FIG. 66. The bevel 394 reduces stress in the housing 104 during operation and helps direct fluid into the discharge bore 178.

(402) Continuing with FIGS. 65 and 68, the second bevel 382 and the second beveled surface 384 are shown in more detail. The second beveled surface 382 is positioned between the second seal 376 and the intake bores 170 and 172. The second beveled surface 384 has an annular shape and surrounds the horizontal bore 106 in an endless manner. In alternative embodiments, the second beveled surface may have a shape that conforms to the shape of the second bevel formed in the fluid routing plug.

(403) When the fluid routing plug 116 is installed within the horizontal bore 106, the second bevel 382 seats against the second beveled surface 384, as shown in FIG. 68. The bevels 382 and 384 meet at a non-right angle. Such angle reduces stress in the fluid routing plug 116 and the housing 104 during operation. The bevels 382 and 384 remain engaged during the forward and backwards stroke of the plunger 290.

(404) Turning to FIGS. 65 and 69, the first bevel 378 and the first beveled surface 380 are shown in more detail. The first beveled surface 380 is positioned between the intake bores 170 and 172 and the first seal 374. The first beveled surface 380 has an annular shape and surrounds the horizontal bore 106 in an endless manner. In alternative embodiments, the first beveled surface may have a shape that conforms to the shape of the first bevel formed in the fluid routing plug.

(405) In contrast to the second bevel 382, the first bevel 378 is sized to be spaced from the first beveled surface 380 when the fluid routing plug 116 is initially installed within the housing 104, as shown by a gap 398. The gap 398 provides space for the fluid routing plug 116 to expand during operation.

(406) As the plunger 290 retracts backwards away from the housing 104, a significant amount of load is applied to the second bevel 382. The applied load causes the fluid routing plug 116 to slightly compress, forcing the intermediate surface 322 at the first bevel 378 to expand outwards. As the first bevel 378 expands, it eventually engages with the first beveled surface 380. Upon engaging the first beveled surface 380, the load being applied to the second bevel 382 is shared with the first bevel 378, thereby decreasing the load applied to the second bevel 382. Without the gap 398, the fluid routing plug 116 would not have room to expand, potentially causing damage to the fluid routing plug 116 and the housing 104 over time.

(407) As the plunger 290 extends forward into the housing 104, the first bevel 378 will return to its un-expanded state, re-creating the gap 398. The gap 398 will repeatedly be created and closed during operation as the plunger 290 reciprocates. In addition to providing space for the fluid routing plug 116 to expand, the gap 398 also provides a gas and fluid relief area during the forward stroke of the plunger 290.

(408) Continuing with FIGS. 68 and 69, because the second bevel 382 carries the majority of the load experienced by the fluid routing plug 116 during operation, the second bevel 382 is longer than the first bevel 378. In alternative embodiments, the first bevel may be longer than that shown in FIG. 69 or be equal in length to the second bevel. In such embodiments, the first beveled surface formed in the housing may correspond with the chosen size of the first bevel. In further alternative embodiments, the first bevel may be sized to mate with the first beveled surface when the fluid routing plug is first installed within the housing.

(409) With reference to FIGS. 65, 70, and 71, in order to prevent fluid from leaking around the fluid routing plug 116 during operation, the first and second seals 374 and 376 are positioned between the sealing surfaces 370 and 372 and the walls of the housing 104 surrounding the horizontal bore 106.

(410) The first seal 374 is positioned within a first annular groove 400 formed in housing 104 and

surrounding the horizontal bore **106** in an endless manner. The first groove **400** is positioned between the intake bores **170** and **172** and the second outer surface **110** of the housing **104**, as shown in FIG. **65**. The first groove **400** is characterized by two sidewalls **402** joined by a base **404**, as shown in FIG. **71**. The sidewalls **402** may join the base **404** via radius corners or at a 90 degree angle. In alternative embodiments, the first groove may have a non-concentric shape that corresponds with the shape of the first sealing surface.

(411) The second seal **376** is positioned within a second annular groove **406** formed in the housing **104** and surrounding the horizontal bore **106** in an endless manner. The second groove **406** is positioned between the discharge bore **178** and the intake bores **170** and **172**, as shown in FIG. **65**. The second groove **406** is characterized by two sidewalls **408** joined by a base **410**. The sidewalls **408** may join the base **410** via radius corners or at a 90 degree angle. In alternative embodiments, the second groove may have a non-concentric shape that corresponds with the shape of the second sealing surface.

(412) The second groove **406** has a larger diameter than that of the first groove **400** due to the diameter of the horizontal bore **106** at each groove, as shown in FIG. **65**. Likewise, the second seal **376** has a larger diameter than that of the first seal **374**. Because the first and second grooves **400** and **406** are formed in the housing **104**, no grooves are formed in the intermediate surface **322** of the fluid routing plug **116** for receiving a seal.

(413) When the fluid routing plug **116** is installed within the horizontal bore **106**, the first and second seals **374** and **376** tightly engage the corresponding first and second sealing surfaces **370** and **372**, as shown in FIGS. **70** and **71**. During operation, the first and second seals **374** and **376** wear against the first and second sealing surfaces **370** and **372**. If the first or second sealing surface **370** or **372** begins to erode, allowing fluid to leak around the fluid routing plug **116**, the plug **116** may be removed and replaced with a new plug **116**. The first or second seal **374** or **376** may also be replaced with a new seal, if needed.

(414) The first groove **400** shown in FIG. **71** is wider than the second groove **406** shown in FIG. **70**. As described below, each groove **400** and **406** is sized to correspond with the size of the seal installed within the groove. In alternative embodiments, the first and second grooves may be wider or narrower than those shown in the figures in order to accommodate the size of the seal installed within the groove.

(415) As discussed above, the fluid routing plug **116** may repeatedly stretch and contract in response to the changing fluid pressure. For example, when the plunger **290** is retracted out of the housing **104**, the fluid pressure at the first surface **318** is equal or approximately equal to the pressure of fluid delivered to the housing **104** from the intake manifolds **166** and **168**. Such fluid pressure may be around 100-200 psi, for example. When the plunger **290** extends into the housing **104**, the fluid at the first surface **318** may be pressurized to around 10,000 psi, for example.

(416) The first seal **374**, being positioned adjacent the first surface **318** of the fluid routing plug **116** experiences the constant change in fluid pressure. In contrast, the second seal **376**, being positioned adjacent the second surface **320**, experiences more static fluid pressure. The fluid pressure at the second surface **320** of the fluid routing plug **116** may remain at or close to 10,000 psi, for example.

(417) Continuing with FIGS. **70** and **71**, because the first seal **374** experiences more pressure fluctuations during operation than the second seal **376**, the first seal **374** may be more robust than the second seal **376**. For example, the first seal **374** is larger than the second seal **376** and has a generally square cross-sectional shape, while the second seal **376** has a circular cross-sectional shape. The first seal **374** may also have a higher durometer value than the second seal **376**. As described below, both seals **374** and **376** are bi-directional seals. In alternative embodiments, the second seal may be of the same construction as the first seal.

(418) Continuing with FIG. **71**, the first seal **374** is shown engaged with both side walls **402** of the first groove **400**. In operation, as the plunger **290** extends into the housing **104**, pressurized fluid pushes against the right side of the first seal **374**, helping to activate the first seal **374** and create a

tight seal between the first seal **374** and the first sealing surface **370**. As the plunger **290** retracts from the housing **104** and the fluid pressure drops, the fluid pressure is greater on the left side of the first seal **374**. Thus, the fluid pressure may push against the left side of the first seal **374**, helping to activate the first seal **374**. Therefore, in operation, the first seal **374** may move slightly between its left and right side.

(419) Continuing with FIG. **70**, the second seal **376** is shown engaged with both side walls **408** of the second groove **406**. In operation, pressurized fluid within the housing **104** helps to activate the second seal **376**, thereby creating a tight seal between the second seal **376** and the second sealing surface **372**. Because the second seal **376** experiences primarily static fluid pressure, the second seal **376** may not move within the second groove **406**, as much as the first seal **374** moves within the first groove **400**.

(420) Continuing with FIGS. **70** and **71**, the first seal **374** also takes up approximately 97% of the open volume within the first groove **400**. Likewise, the second seal **376** takes up almost 97% of the open volume within the second groove **406**. Normally, seals are configured to take up around 70% of the open volume within the groove the seal is installed within. The remaining open volume provides space for the seal to expand and move. However, in operation, fluid and proppants can fill the open volume and wear against the groove, eventually causing the walls of the groove to erode. If the walls of the groove are damaged, the housing **104** may need to be replaced.

(421) By sizing the grooves **400** and **406** so that the seals **374** and **376** take up almost all of the open volume within the corresponding grooves **400** and **406**, there is less room for fluid or proppants to fill any open space within the grooves. Specifically, fluid and proppants are prevented from entering any open volume on the back side of the seals **374** and **376**, thereby protecting the first and second grooves **400** and **406** from erosion. In alternative embodiments, the first seal may take less volume of the first groove than is shown in FIG. **70**. Likewise, in alternative embodiments, the second seal may take up less volume of the second groove than is shown in FIG. **71**. The other grooves formed in the housing and described herein may also be configured so that the corresponding seals take up approximately 97% of the open volume within the groove.

(422) Continuing with FIG. **71**, the first sealing surface **370** may extend up to immediately adjacent the first surface **318** of the body **316**. A first portion **412** of the intermediate surface **322** between the first bevel **378** and the first sealing surface **370** faces the housing **104** walls. A very small gap exists between the first portion **412** and the housing **104**. The gap may be as small as 0.001 inches in width. Such gap provides clearance to reduce friction between the fluid routing plug **116** and the housing **104** during installation and operation. Such gap also provides space for excess proppant to collect during operation.

(423) Continuing with FIG. **70**, a second portion **416** of the intermediate surface **322** between the second sealing surface **372** and the second surface **320** may face the walls of the housing **104**. A third portion **418** of the intermediate surface **322** between the second sealing surface **372** and the transition bevel **386** may also face the walls of the housing **104**. Like the first portion **412**, a very small gap exists between the second and third portions **416** and **418** and the housing **104**. The gaps may be as small as 0.001 inches in width. Such gaps provide clearance to reduce friction between the fluid routing plug **116** and the housing **104** during installation and operation. Such gaps also provide space for excess proppant to collect during operation.

(424) Turning back to FIG. **65**, as discussed above, the walls of the housing **104** surrounding the horizontal bore **106** are sized to allow for easy installation of the fluid routing plug **116**. The second groove **406** has a diameter **D1**. A maximum diameter of the second beveled surface **384** has a diameter **D2**. A maximum diameter of the first beveled surface **380** has a diameter **D3**, and the first groove **400** has a diameter **D4**. The diameter **D4** is greater than the diameter **D3**. The diameter **D3** is greater than the diameter **D2**, and the diameter **D2** is greater than the diameter **D1**.

(425) With reference to FIGS. **72-76** and **85-89**, the suction and discharge valves **292** and **294** are generally identical, with the exception that the discharge valve **294** may be larger in size than the

suction valve **292**. As discussed above, the suction and discharge valves **292** and **294** each have a sealing element **346** and **360**. The sealing elements **346** and **360** each include a sealing surface **420** and **422** that tapers at an angle that matches the angle of the tapered wall **340** and **356** of the fluid routing plug **116**. Thus, the sealing surfaces **420** and **422** each taper at an angle of 30 or 45 degrees. Preferably, the tapered walls **340** and **356** and the sealing surfaces **420** and **422** both taper at an angle of 45 degrees.

(426) Forming the mating tapered walls **340** and **356** and sealing surfaces **420** and **422** at 45 degrees provides more surface area for the valves **292** and **294** to seal against the fluid routing plug **116**. Providing more sealing surface area or a larger “strike face” helps distribute the forces applied to the valves **292** and **294** and the fluid routing plug **116**, thereby providing more evenly distributed sealing. Providing more evenly distributed sealing prevents certain areas from wearing faster than others, helping to increase the life of the parts.

(427) Each valve **292** and **294** also has an outer sealing diameter A and an inner sealing diameter B, as shown in FIGS. **72** and **85**. The ratio of the outer sealing diameter A to the inner sealing diameter B is preferably 1.55 or greater. This ratio helps increase the life of the valves **292** and **294** and reduce any turbulent fluid flow during operation. The valves **292** and **294** and the fluid routing plug **116** are configured so that no portion of the valves **292** and **294** enters the first or second fluid passages **326** and **336** during operation. Additionally, no portion of the valve **292** enters the blind bore **328** during operation. Rather, the suction valve **292** is configured only to cover the entrance **342** of the blind bore **330** on the first surface **318**, and the discharge valve **294** is configured only to cover the openings **364** of the second fluid passages **336** on the second surface **320**.

(428) Continuing with FIGS. **72-76**, the suction valve **292** is shown in more detail. The suction valve **292** comprises the sealing element **346** joined to a stem **424**. When the suction valve **292** is installed within the horizontal bore **106**, the stem **424** extends along an axis that is parallel to or aligned with central the longitudinal axis **114** of the housing **104**.

(429) The sealing element **346** comprises opposed first and second surfaces **426** and **428** joined by the sealing surface **420**. A groove **430** is formed in the sealing surface **420** adjacent the first surface **426**, as shown in FIG. **76**. A seal **432** is installed within the groove **430**. The groove **430** is characterized by a first sidewall **434** joined to a second sidewall **436**. The sidewalls **434** and **436** may be joined by an inner groove **438**. The groove **430** is sized to correspond with the inward facing surface of the seal **432**. An outward facing surface of the seal **432** comprises a convex surface **440** joined to a concave surface **442**. The seal **432** is preferably made of a polyurethane compound. In alternative embodiments, the seal may be made of a different elastomer material.

(430) When the suction valve **292** seals against the first surface **318** of the fluid routing plug **116**, the seal **432** and a portion of the sealing surface **420** mate with the tapered wall **340**, as shown in FIG. **51**. The seal **432** is shaped so that the convex surface **440** displaces into, or toward, the concave surface **442** as the seal **432** engages the tapered wall **340**. This relative movement allows the shear forces to be dissipated, increasing the life of the seal **432** and the suction valve **292**. If the seal **432** becomes worn and no longer seals properly, the seal **432** may be removed and replaced with a new seal **432**. In alternative embodiments, the seal and groove may have various shapes and sizes, as desired. In further alternative embodiments, the sealing surface may not include a groove and corresponding seal.

(431) Continuing with FIG. **76**, the second surface **428** of the sealing element **346** is sized to cover the entrance **342** of the blind bore **328**, as shown in FIG. **51**. A cutout **444** is formed within the second surface **428**. The cutout **444** creates a small cavity within the second surface **428**. The cavity provides space for fluid to collect and apply pressure to the suction valve **292**. Such pressure helps force the suction valve **292** to move axially to an open position.

(432) Continuing with FIGS. **75** and **76**, the stem **424** projects from the first surface **426** of the sealing element **346**. An annular void **446** is formed in the first surface **426** and surrounds the stem **424**. The first surface **426** further includes a ring-shaped outer rim **448** that surrounds the annular

void **446** and the stem **424**. The outer rim **448** joins the sealing surface **420**. The annular void **446** reduces weight within the suction valve **292** and helps orient the valve's center of gravity during operation.

(433) An annular groove **450** is formed in the outer rim **448**. The groove **450** is configured for receiving a bottom portion of a spring **452**, as shown in FIG. **83**. As described below, a top portion of the spring **452** engages with the suction valve guide **296**, as shown in FIGS. **83** and **84**. The spring **452** biases the suction valve **292** in the closed position. Positioning the spring **452** on the outer rim **448** helps to stabilize the suction valve **292** during operation.

(434) With reference to FIGS. **77-82**, the stem **424** is configured to move axially within the suction valve guide **296**. The suction valve guide **296** may also be referred to as a cage for the suction valve **292**. The suction valve guide **296** comprises a body **454** having opposed first and second surfaces **456** and **458**. A central passage **460** is formed within the body **454** and interconnects the first and second surfaces **456** and **458**. A plurality of legs **462** extend out from the body **454** adjacent its first surface **456** and project downward towards its second surface **458**. The suction valve guide **296** shown in FIGS. **77-82** has six evenly spaced legs **462** formed around its body **454**. In alternative embodiments, more or less than six legs may be formed on the body and may be non-uniformly spaced.

(435) The legs **462** gradually decrease in thickness from the body **454** to a bottom surface **464** of each leg **462**. The bottom surface **464** of each leg **462** is extremely thin so that the legs **462** do not block or interfere with the openings **348** of the second fluid passages **336** on the first surface **318**, as shown in FIG. **50**.

(436) Continuing with FIGS. **77-82**, an outer surface of each leg **462** includes a bevel **466**. The bevels **466** are configured to engage a corresponding bevel **468** formed in the walls of the housing **104**, as shown in FIGS. **50** and **51**. The suction valve guide **296** is inserted into the horizontal bore **106** until the bevels **466** and **468** engage, allowing the guide **296** to bottom out on the walls of the housing **104**. Once the bevels **466** and **468** are engaged, the suction valve guide **296** is held against the walls of the housing **104** by the spring **452** and fluid pressure.

(437) When the suction valve guide **296** is in its installed position, the bottom surface **464** of each of the legs **462** hovers just above the first surface **318** of the fluid routing plug **116**, leaving a gap between the legs **462** and the plug **116**. The bottom surfaces **464** do not directly contact the fluid routing plug **116** in order to prevent the suction valve guide **296** from applying load to the plug **116** during operation.

(438) Continuing with FIG. **80**, a tubular insert **470** is installed within the central passage **460** of the body **454**. The insert **470** may be press-fit within the passage **460**. The insert **470** extends the length of the central passage **460** and is formed from a more wear resistant material than the suction valve guide **296**. For example, the insert **470** may be made of tungsten carbide, while the suction valve guide **296** may be made of high strength alloy steel. The stem **424** is installed within the insert **470** and reciprocates within the insert **470** during operation, as shown in FIGS. **50** and **51**. Any fluid contained within the insert **470** drains from the opening of the central passage **460** on the first surface **456** of the body **454**.

(439) During operation, the stem **424** may wear against the insert **470** as it reciprocates. The insert **470** helps decrease the rate of wear and helps the stem to wear evenly against the insert. Forming only the insert **470** out of a wear resistant material helps reduce the cost of the other parts, which do not experience as much wear during operation.

(440) Turning to FIGS. **83** and **84**, the spring **452** is interposed between the suction valve **292** and the suction valve guide **296**. The spring **452** is held between the outer rim **448** of the suction valve **292** and an inner surface **472** of the legs **462**. At least a portion of the spring **452** surrounds the body **454** of the suction valve guide **296**. As the suction valve **292** moves to an open position, the spring **452** compresses between the suction valve **292** and the suction valve guide **296**.

(441) With reference to FIGS. **85-89**, the discharge valve **294** is shown in more detail. As discussed

above, the discharge valve **294** is constructed identically to the suction valve **292**, with the exception that the discharge valve **294** may be larger in size. The discharge valve **294** shown in FIGS. **85-89**, for example, has a larger sealing surface **422** and a longer stem **474** than the suction valve **292**. When the discharge valve **294** is installed within the horizontal bore **106**, the stem **424** extends along an axis that is parallel to or aligned with the central longitudinal axis **114** of the housing **104**. A seal **475** is installed within a groove **477** formed in the sealing surface **422** and is configured to engage with the tapered wall **356** formed in the second surface **320** of the fluid routing plug **116**. A bottom surface **476** of the discharge valve **294** is sized to cover the central base **354**, as shown in FIG. **50**.

(442) With reference to FIGS. **90-95**, the stem **474** formed on the discharge valve **294** is configured to move axially within the discharge valve guide **298**. The discharge valve guide **298** may also be referred to as a cage for the discharge valve **294**. The discharge valve guide **298** comprises a body **478** having opposed first and second surfaces **480** and **482** joined by an intermediate surface **484**. The intermediate surface **484** includes a front portion **486**, a medial portion **488**, and a rear portion **490**. The medial portion **488** has a larger diameter than both the front and rear portions **486** and **490**. The front portion **486** has a slightly larger diameter than the rear portion **490**.

(443) Continuing with FIGS. **90-95**, a blind bore **492** is formed in the first surface **480** and extends into the front portion **486** of the body **478**. The blind bore **492** is configured to receive a tool used to grip the discharge valve guide **298**. The front portion **486** is sized to be received within the cutout **308** formed in the retainer **300**, as shown in FIGS. **50** and **51**. When the discharge valve guide **298** and the retainer **300** are engaged, the blind bore **492** opens into the central passage **310** formed in the retainer **300**.

(444) A central passage **494** is formed in the body **478** and opens on the second surface **482**, as shown in FIGS. **93** and **94**. The central passage **494** opens in the body **478** into an axially blind counterbore **496**. A plurality of relief ports **498** are formed in the body **478**. Each relief port **498** interconnects the counterbore **496** and a base **500** of the medial portion **488**, as shown in FIG. **94**.

(445) Continuing with FIGS. **93** and **94**, a tubular insert **502** is installed within the central passage **494**. The insert **502** is identical to the insert **470**, with the exception that the insert **502** may be larger than the insert **470**. During operation, the stem **474** moves axially within the insert **502** installed within the central passage **494**. Any fluid within the insert **502** drains from the body **478** through the counterbore **496** and the relief ports **498**.

(446) Continuing with FIGS. **90-92**, a plurality of legs **504** project from the medial portion **488** and extend towards the second surface **482** of the body **478**. The discharge valve guide **298** shown in FIGS. **90-95** comprises five legs **504**. The legs **504** are positioned on the body **478** so as to leave a large space **506** between at least two adjacent legs **504**. Other than the space **506**, the legs **504** are equally spaced from one another. The space **506** is intended to align with the discharge bore **178**, thereby preventing any legs **504** from blocking the discharge bore **178** during operation. Providing the space **506** therefore allows fluid to flow freely between the discharge valve **294** and the discharge bore **178** without significant obstructions. The space **506** also helps minimize wear applied to the legs **504** by the flowing fluid over time. In alternative embodiments, the body may have more or less than five legs and be spaced, as desired, as long as the legs are positioned on the body so as to leave a large space between at least two of the legs.

(447) With reference to FIG. **90**, each of the legs **504** has a thicker upper portion **508** and thinner lower portion **510**. The thicker upper portion **508** provides strength to the legs **504** while the lower portion **510** is thinned in order to provide more room for fluid flow around the legs **504**. The upper portion **508** also includes a tapered inner surface **512**. Tapering the inner surface **512** of the legs **504** provides strength and alleviates stress in the legs **504** during operation.

(448) Continuing with FIGS. **90-95**, when the discharge valve guide **298** is installed within the horizontal bore **106**, a bottom surface **514** of each leg **504** engages the outer rim **352** of the second surface **320** of the fluid routing plug **116**, as shown in FIGS. **50** and **51**. The discharge valve guide

298 is held against the fluid routing plug **116** by the retainer **300**. Such engagement helps keep the second bevel **382** of the fluid routing plug **116** seated against the second beveled surface **384**, as shown in FIGS. **65** and **68**.

(449) Continuing with FIGS. **93** and **95**, a dowel pin **516** is installed within a blind bore **517** formed in the medial portion **488** of the body **478**. The dowel pin **516** is configured to be received within a dowel pin hole or groove **518** formed in the walls of the housing **104** surrounding the horizontal bore **106**, as shown in FIGS. **96** and **97**. The discharge valve guide **298** is installed within the horizontal bore **106** such that the dowel pin **516** is positioned within the dowel pin hole **518**. Such positioning ensures that the space **506** between the pair of legs **504** aligns with the discharge bore **178**, thus preventing any legs **504** from blocking the discharge bore **178** during operation.

(450) Continuing with FIGS. **96** and **97**, a seal **520** is interposed between the intermediate surface **484** of the body **478** and the walls of the housing **104**. The seal **520** may be identical to the second seal **376** shown in FIGS. **65** and **70**. In alternative embodiments, the seal may be identical to the first seal **374** shown in FIGS. **65** and **71**. The seal **520** is installed within a groove **522** formed in the housing **104**. The groove **522** is characterized by two sidewalls **524** joined to a base **526**. The sidewalls **524** may join the base **526** via radius corner or at a 90 degree angle. During operation, the seal **520** wears against the outer intermediate surface **484** of the discharge valve guide **298**. If the intermediate surface **484** begins to erode, allowing fluid to leak around the seal **520**, the discharge valve guide **298** may be removed and replaced with a new discharge valve guide **298**.

(451) With reference to FIGS. **98** and **99**, a spring **528** is installed between the discharge valve **294** and the discharge valve guide **298**. A bottom portion of the spring **528** sits in a groove **530**, shown in FIG. **89**, formed in an outer rim **532** of the discharge valve **294**. A top portion of the spring **528** engages a ledge **534** formed in the base **500** of the medial portion **488** of the discharge valve guide **298**. During operation, the spring **528** compresses against the ledge **534** of the medial portion **488**.

(452) Turning to FIG. **100**, the components installed within the housing **104** are installed through the first surface **108**, starting with the suction valve guide **296**. The diameter of the installed components slightly increases from the second surface **320** to the first surface **318**. For example, the suction valve guide **296** has smaller outer diameters than the fluid routing plug **116**, and the fluid routing plug **116** has smaller outer diameters than the discharge valve guide **298**. The discharge valve guide **298** has smaller outer diameters than the retainer **300**.

(453) Likewise, the diameters of the walls surrounding the horizontal bore **106** generally increase from the second surface **110** to the first surface **108**. As shown in FIG. **100**, a diameter **D4** of the horizontal bore **106** is greater than a diameter **D3** of the horizontal bore **106**. The diameter **D3** of the horizontal bore **106** is greater than a diameter **D2** of the horizontal bore **106**. The diameter **D2** of the horizontal bore **106** is greater than a diameter **D1** of the horizontal bore **106**. Such construction allows the components to be installed without engaging the walls of the housing **104** until the component is at its intended installed position. The seals **374**, **376**, and **520** may be installed within the housing **104** prior to installing the other components described above.

(454) Turning to FIGS. **100A-100E**, another embodiment of a fluid routing plug **550** is shown. The fluid routing plug **550** may be installed within the housing **104** in place of the fluid routing plug **116**. The fluid routing plug **550** is identical to the fluid routing plug **116**, with a few exceptions. The fluid routing plug **550** comprises a body **552** having a first outer surface **554** joined to a second outer surface **556** by an intermediate outer surface **558**. The second surface **556** of the fluid routing plug **550** is generally identical to the second surface **320** of the fluid routing plug **116**, but a central base **560** formed in the second surface **556** is spaced from an edge **562** of a tapered wall **564** formed in the second surface **556**. The central base **560** is spaced from the tapered wall **564** such that a throat **566** is formed between the central base **560** and the tapered wall **564**.

(455) Continuing with FIGS. **100A-100D**, a blind hole **568** is formed in the central base **560** and a plurality of openings **570** corresponding to a plurality of second fluid passages **572** open on the

central base **560** and surround the blind hole **568**. In operation, fluid exiting the openings **570** flows into the throat **566** before pushing against the discharge valve **294** engaged with the second surface **556**. Allowing fluid to gather in the throat **566** before contacting the discharge valve **294** helps the fluid to contact more surface area of the discharge valve **294**, instead of having a plurality of single points of contact from each second fluid passage opening. Allowing the fluid to contact more surface area of the discharge valve **294** helps reduce wear to the valve over time.

(456) Continuing with FIGS. **100E** and **100F**, the intermediate surface **558** of the fluid routing plug **550** is identical to the intermediate surface **322** formed on the fluid routing plug **116**. However, the intermediate surface **558** may include a cutout **576** adjacent the second surface **556**. The cutout **576** provides space for fluid or proppant to collect during operation, as shown in FIG. **100F**. The cutout **576** also helps reduce friction during installation of the fluid routing plug **550** within the housing **104**. A small gap **578** may also exist between the walls of the housing **104** and the intermediate surface **558** between a second sealing surface **580** and the cutout **576**, as shown in FIG. **100F**. The gap **578** helps the seal **376** breath during operation.

(457) Continuing with FIG. **100B**, the first surface **554** of the fluid routing plug **550** is identical of the first surface **318** of the fluid routing plug **116**, with the exception of its outer rim **582**. The outer rim **582** is flat and wider than the outer rim **338**, shown in FIG. **55**. Because the outer rim **582** is wider, a plurality of openings **584** for the second fluid passages **572** may have a slightly larger diameter than the openings **348**, shown in FIG. **56**. Likewise, the openings **570** may also have a slightly larger diameter than the openings **364** shown in FIG. **54**. Providing a slightly larger diameter for the second fluid passages **572** helps reduce fluid velocity through the fluid routing plug **550** during operation. Reducing fluid velocity within the fluid routing plug **550** helps reduce wear to the fluid routing plug **550** over time.

(458) Turning to FIGS. **101-109**, another embodiment of a fluid routing plug **600** is shown. The fluid routing plug **600** may be installed within the housing **104** in place of the fluid routing plug **116**. The fluid routing plug **600** is identical to the fluid routing plug **116**, with a few exceptions. The fluid routing plug **600** comprises a body **602** having a first outer surface **604** joined to a second outer surface **606** by an intermediate outer surface **608**. In contrast to the fluid routing plug **116**, the first and second surfaces **604** and **606** of the fluid routing plug **600** are configured so that each surface **604** and **606** has identically sized tapered walls **610** and **612**, as shown in FIG. **103**. Because the tapered walls **610** and **612** are the same size, a suction valve **614** and a discharge valve **616** used with the fluid routing plug **600** may be identical in size, as shown in FIGS. **108** and **109**.

(459) Using the same size suction and discharge valves **614** and **616** helps equalize the forces applied to the fluid routing plug **600** and the valves **614** and **616** during operation, helping to reduce any wear to the parts over time. Making the suction and discharge valves **614** and **616** identical also makes replacing the valves **614** and **616** during operation easier.

(460) Continuing with FIGS. **103**, **105**, and **106**, the tapered wall **612** formed in the second surface **606** extends between an outer rim **618** and an annular groove **620** formed in the center of the second surface **606**. The annular groove **620** may be considered a central base formed in the second surface **606**. The groove **620** surrounds a blind bore **622** formed in the center of the second surface **606**. The blind bore **622** is identical to the blind bore **362** formed in the fluid routing plug **116**, as shown in FIG. **55**.

(461) The groove **620** is characterized by two parallel sidewalls **624** joined by a base **626**. The sidewalls **624** each extend at a non-zero angle relative to a central longitudinal axis **628** of the body **602**. Because the sidewalls **624** of the groove **620** extend at an angle, the base **626** of the groove **620** extends at a non-zero angle relative to the central longitudinal axis **628** of the body **602**. Preferably, the base **626** extends at approximately the same angle as the tapered wall **612** so that the base **626** and the tapered wall **612** are in a generally parallel relationship. The tapered wall **612** shown in FIG. **103** extends at a 45 degree angle relative to the central longitudinal axis **628**.

(462) An annular inner edge **638** of the tapered wall **612** is joined to the outer sidewall **624** of the

groove **620** at a right angle. The diameter of the inner edge **638** of the tapered wall **612** is the same size as a diameter of an entrance **630** of an axially blind bore **632** formed in the first surface **604**, as shown in FIG. **103**. In alternative embodiments, the groove formed in the second surface and the inner edge of the tapered wall may not have an annular shape.

(463) Continuing with FIGS. **103**, **105**, and **106**, a plurality of second fluid passages **634** are formed in the body **602**. The second fluid passages **634** are identical to the second fluid passages **336** formed in the fluid routing plug **116**, shown in FIGS. **52-64**, with the exception of the positioning of their openings **636** on the second surface **606**. Each second fluid passage **634** opens on the base **626** of the groove **620** formed in the second surface **606**. Thus, the openings **636** are axially spaced from the inner edge **638** of the tapered wall **612**. Because the sidewalls **624** of the groove **620** are formed at an angle, the inner edge **638** of the tapered wall **612** slightly overlaps the openings **636**, as shown in FIG. **106**. By positioning the openings **636** in an axially spaced relationship with the inner edge **638** of the tapered wall **612**, the size of the tapered wall **612** can be decreased without decreasing the size of the openings **636**. The annular groove **620** also functions as a throat, similar to the throat **566** formed in the fluid routing plug **550**.

(464) Because the tapered wall **612** is decreased in size from the tapered wall **356** shown in FIG. **55**, the outer rim **618** on the second surface **606** is wider than the outer rim **352**. The outer rim **618** also tapers between the intermediate surface **608** and the tapered wall **612**, as shown in FIGS. **103** and **104**. Such taper increases the length of the tapered wall **612** without increasing the length of the intermediate surface **608**.

(465) Continuing with FIGS. **101-107**, the first surface **604** is identical to the first surface **318** shown in FIGS. **53**, **55**, and **56**, with the exception of its outer rim **640**. Instead of tapering like the outer rim **338**, shown in FIG. **55**, the outer rim **640** is flat. The outer rim **640** is flat in order to slightly decrease the size of the tapered wall **610** to match the size of the tapered wall **612**. The intermediate surface **608** of the fluid routing plug **600** is identical to that of the fluid routing plug **116**, shown in FIG. **64**. A plurality of first fluid passages **642** formed in the body **602** are identical to the first fluid passages **326**, shown in FIGS. **55**, **57**, and **59**. The second fluid passages **634** open on the outer rim **640** of the first surface **604**, as shown by the openings **644**. The openings **644** are positioned in groups **645**, in the same manner as the second fluid passages **336** formed in the fluid routing plug **116**, as shown in FIG. **56**. The openings **636** on the second surface **606** may remain spaced in groups **645**, as shown in FIG. **106**.

(466) With reference to FIGS. **108** and **109**, the fluid routing plug **600** routes fluid throughout the housing **104** in the same manner as the fluid routing plug **116**. The suction valve guide **296** is shown engaged with suction valve **614**. Another embodiment of a discharge valve guide **647** is shown engaged with the discharge valve **616**.

(467) The discharge valve guide **647** is identical to the discharge valve guide **298**, shown in FIGS. **90-95**, with a few exceptions. A counterbore **649** formed in the guide **647** is larger than the counterbore **496**. The counterbore **649** is larger in order to accommodate the shorter stem **646** of the discharge valve **616**. An insert **651** installed within the discharge valve guide **647** is the same size as the insert **470** installed within the suction valve guide **296**.

(468) With reference to FIGS. **110-114**, as discussed above, in contrast to the valves **292** and **294**, the valves **614** and **616** are identical in size and shape. The valves **614** and **616** are generally identical to the valves **292** and **294**, with a few exceptions. Each valve **614** and **616** comprises a sealing element **652** joined to a stem **646**. The stem **646** projects from a first surface **650** of the sealing element **652**.

(469) An annular cutout **648** is formed within a medial portion of the stem **646**. The cutout **648** provides space for fluid or proppants to collect during operation. Providing such space prevents the fluid and proppants from rubbing against the inserts **470** and **502**. The suction and discharge valves **292** and **294** may be configured to include an annular cutout within their stems **424** and **474**.

(470) Continuing with FIGS. **110-114**, the sealing element **652** further includes a second surface

668 joined to the first surface **650** by a sealing surface **658**. A groove **656** is formed in the sealing surface **658** for housing a seal **654**. The groove **656** is identical to the groove **430**, shown in FIG. **76**. An outward facing surface of the seal **654** comprises a sidewall **660** joined to a tapered base **662**. In operation, the tapered base **662** engages the tapered walls **610** and **612** of the fluid routing plug **600**. The sidewall **660** may compress creating a tight seal.

(471) The first surface **650** of the sealing element **652** includes an outer rim **664**. An outer ledge **666** surrounds the outer rim **664**. A bottom portion of a spring engages the outer rim **664** and is held in place by the outer ledge **666**. While not shown, a cutout may be formed in the second surface **668** of the sealing element **652**, like the cutout **444**, shown in FIG. **76**.

(472) One or more kits may be useful in assembling the fluid end section **102**. A kit may comprise a plurality of housings **104** and a plurality of the corresponding fluid routing plugs **116** or **600**. The kit may also comprise a plurality of suction valves **292** or **614**, discharge valves **294** or **616**, suction valve guides **296**, discharge valve guides **298** or **657**, springs **452** and **528**, retainer **300**, stuffing box **140**, retainer **232**, plunger packing **224**, packing nut **276**, fastening system **234**, discharge conduit **174**, and the various seals described herein. The kit may also comprise the intake manifolds **166** and **168**, pipe system **176**, connect plate **118**, fastening system **146** and stay rods **120**. The kit may also comprise other various features described herein for use with the fluid end **100**. Unless specifically described herein, the various components of the fluid end **100** may be made of high strength alloy steel, such as carbon steel or stainless steel.

(473) With reference to FIGS. **115-117**, an alternative embodiment of a fluid routing plug **700** is shown. The fluid routing plug **700** may be installed within the housing **104** in place of the fluid routing plug **116**. The fluid routing plug **700** is identical to the fluid routing plug **116**, with the exception of the shape of its first and second bevels **702** and **704**. When the fluid routing plug **700** is first installed within the horizontal bore **106**, the second bevel **704** only partially engages a second beveled surface **706**, as shown in FIG. **116**. The bevels **704** and **706** mate at a second bevel mating surface **708** and a second beveled surface mating surface **710**. Below the mating surfaces **708** and **710**, the second bevel **704** and the second beveled surface **706** have mating angles that are not equal, causing a gap **712** to exist between the bevels **704** and **706**. Specifically, the second bevel **704** may have a slightly convex shape so that portions of the second bevel **704** don't match the flat shape of the second beveled surface **706**.

(474) The width of the gap **712** gradually increases between the mating surfaces **708** and **710** and a bottom portion **714** of the second bevel **704** and a bottom portion **716** of the second beveled surface **706**. Thus, the width B of the gap **712** is wider than the width A of the gap **712**. Because the second bevel **704** has a slightly convex shape, the angle between the mating surfaces **708** and **710** is different from the angle between the bottom portions **714** and **716**.

(475) Turning to FIG. **117**, the first bevel **702** and the first beveled surface **718** are shown in more detail. Like the second bevel **704**, the first bevel **702** may only partially engage a first beveled surface **718**. The bevels **702** and **718** mate at a first bevel mating surface **720** and a first beveled surface mating surface **722**. Below the mating surfaces **720** and **722**, the first bevel **702** and the first beveled surface **718** have mating angles that are not equal, causing a gap **724** to exist between the bevels **702** and **718**. Specifically, the first bevel **702** may have a slightly convex shape so that portions of the first bevel **702** do not match the flat shape of the first beveled surface **718**.

(476) The width of the gap **724** gradually increases between the mating surfaces **720** and **722** and a bottom portion **726** of the first bevel **702** and a bottom portion **728** of the first beveled surface **718**. Thus, the width B of the gap **724** is wider than the width A of the gap **724**. Because the first bevel **702** has a slightly convex shape, the angle between the mating surfaces **720** and **722** is different from the angle between the bottom portions **726** and **728**.

(477) The width of the gaps **712** and **724** has been exaggerated in FIGS. **116** and **117** for illustration purposes. In reality, portions of the gaps **712** and **724** may be approximately 0.002 inches in width, for example. However, the gaps **712** and **724** may be wider or smaller depending on the materials

and forces used.

(478) As discussed above, in operation, the fluid pressure applied to the fluid routing plug **700** will cause the plug **700** to compress and expand as the plunger **290** retracts from the housing **104**. As the fluid routing plug **700** starts to expand, the bottom portion **714** of the second bevel **704** will move to engage the bottom portion **714** of the second beveled surface **706**, causing the bottom portions **714** and **716** to mate. Likewise, the bottom portion **726** of the first bevel **702** will move to engage the bottom portion **728** of the first beveled surface **718**. Such movement of the fluid routing plug **700** distributes the load applied to the fluid routing plug **700** through the length of the first and second bevels **702** and **704**.

(479) With reference to FIGS. **118-120**, an alternative embodiment of a fluid routing plug **800** is shown. The fluid routing plug **800** may be installed within the housing **104** in place of the fluid routing plug **116**. The fluid routing plug **800** is identical to the fluid routing plug **700**, with the exception of the shape of its first and second bevels **802** and **804**. Like the fluid routing plug **700**, the second bevel **804** is sized to leave a gap **806** between the second bevel **804** and a second beveled surface **808** when the fluid routing plug **800** is first installed within the housing **104**. In contrast to the gap **712**, an angle formed between the mating surfaces **810** and **812** and bottom portions **814** and **816** of the second bevel **804** and second beveled surface **808** remains the same. Thus, an area A of the gap **806** has the same angle as an area B of the gap **806**.

(480) Likewise, the first bevel **802** is shaped so that an angle formed between the first bevel **802** and a first beveled surface **820** stays relatively the same between mating surfaces **822** and **824** and bottom portions **826** and **828**. Thus, the width A of the gap **818** has approximately the same angle as the width B of the gap **818**.

(481) The width of the gaps **806** and **818** has been exaggerated in FIGS. **119** and **120** for illustration purposes. In reality, portions of the gaps **806** and **818**, for example, may be approximately 0.002 inches in width. However, the gaps **806** and **818** may be wider or smaller depending on the materials and forces used.

(482) As discussed above, the first and second bevels **802** and **804** expand during operation. Such movement of the fluid routing plug **800** distributes the load applied to the fluid routing plug **800** through the length of the first and second bevels **802** and **804**.

(483) In alternative embodiments, the first bevel may be configured to have a gap that increases in size, as shown in FIG. **117**, while the second bevel may be configured to have a gap that increases by a different amount, as shown in FIG. **119**, and vice versa. In further alternative embodiments, the width of the gap may be of various shapes and sizes depending on the materials used and forces involved. In even further alternative embodiments, the intermediate surface of the fluid routing plug may include any combination of the different bevel constructions described herein.

(484) Turning to FIGS. **121-128**, another embodiment of a fluid routing plug **900** is shown. The fluid routing plug **900** may be installed within the housing **104** in place of the fluid routing plug **116**. The fluid routing plug **900** is identical to the fluid routing plug **116**, with a few exceptions. The fluid routing plug **900** comprises a body **902** having a first outer surface **904** joined to a second outer surface **906** by an intermediate outer surface **908**. A plurality of first fluid passages **910** are formed in the body **902** and interconnect the intermediate surface **908** and the first surface **904** by way of an axially blind bore **912**, as shown in FIG. **124**.

(485) In contrast to the first fluid passages **326**, shown in FIGS. **55** and **58**, a longitudinal axis **914** of each first fluid passage **910** does not intersect a central longitudinal axis **916** of the body **902**, as shown in FIG. **125**. Rather, the first fluid passages **910** are formed such that the longitudinal axis **914** of each passage **910** is offset from the central longitudinal axis **916** of body **902**. The offset configuration of the first fluid passages **910** encourages a vortex type flow of fluid about the central longitudinal axis **916**, thereby reducing fluid turbulence during operation. In alternative embodiments, the longitudinal axis **914** of each first fluid passage **910** may intersect the longitudinal axis **916** of the body **902**.

(486) A plurality of openings **918** formed on the intermediate surface **908** for the first fluid passages **910** are similar to the openings **334**, shown in FIGS. **57** and **59**, but have a more oblong shape, as shown in FIG. **121**. The oblong shape shown in FIG. **121** has opposed first and second ends **920** and **922**. The second end **922**, which is closer to the second surface **906**, is slightly wider than the first end **920**. The unequal size of the ends **920** and **922** helps direct fluid along the offset longitudinal axis **914** of the first fluid passages **910**. The unequal size of the ends **920** and **922** also helps increase the wall thickness in certain areas of the body **902** between the first fluid passages **910** and a plurality of second fluid passages **924**.

(487) In alternative embodiments, the opposed ends of the openings may be identical in size or may be shaped identical to the openings **334**, shown in FIGS. **57** and **59**. The opening **918** of the first fluid passage **910** shown in FIG. **121** extends along an axis that is parallel to the longitudinal axis **916** of the body **902**. In alternative embodiments, the openings of the first fluid passages may extend at a non-zero angle relative to the longitudinal axis **916** of the body **902**, as shown for example by the openings **972** shown in FIG. **128D**. The angle at which the first fluid passages **910** are formed in the body **902** may vary, as desired, in order to increase the wall thickness within the body **902** and reduce stress in the body **902** during operation.

(488) Continuing with FIGS. **122** and **126**, each of the second fluid passages **924** formed in the body **902** interconnects the first and second surfaces **904** and **906**. The second fluid passages **924** are identical to the second fluid passages **336**, shown in FIGS. **60-63**, but the second fluid passages **924** are slightly pivoted from the position of the second fluid passages **336**. Each second fluid passage **924** is pivoted so that it has a compound angle with respect to the central longitudinal axis **916**, as shown in FIGS. **122**, **123**, **126**, and **127**. Meaning, each second fluid passage **924** extends such that it has two different angles relative to the central longitudinal axis **916**—up-and-down, and side-to-side. Like the first fluid passages **910**, forming the second fluid passages **924** at such angles encourages a vortex type flow of fluid about the central longitudinal axis **916**, thereby reducing fluid turbulence during operation.

(489) Continuing with FIGS. **124**, **127**, and **128**, the first surface **904** of the fluid routing plug **900** may be identical to the first surface **318**, shown in FIGS. **53**, **55**, and **56**. However, an outer rim **926** of the first surface **904** may be flat rather than tapered. The second surface **906** of the fluid routing plug **900** is identical to the fluid routing plug **116**, but a central base **928** formed in the second surface **906** may be slightly set back within the body **902**, as compared to the central base **354**, shown in FIGS. **54** and **55**. An outer rim **930** on the second surface **906** may be slightly wider than the outer rim **352**, shown in FIGS. **54** and **55**. The intermediate surface **908** of the fluid routing plug **900** may be identical to the intermediate surface **322** of the fluid routing plug **116**.

Alternatively, the intermediate surface may be identical to those formed on the fluid routing plug **700** or **800**.

(490) In alternative embodiments, the first and second surfaces **904** and **906** of the fluid routing plug **900** may be configured so that its tapered walls **932** and **934** are the same size, like the fluid routing plug **600**. In further alternative embodiments, the first and second surfaces of the fluid routing plug **900** may be identical to the first and second surfaces of the fluid routing plug **116**.

(491) Turning to FIGS. **128A-128G**, another embodiment of a fluid routing plug **950** is shown. The fluid routing plug **950** may be installed within the housing **104** in place of the fluid routing plug **116**. The fluid routing plug **950** is identical to the fluid routing plug **900**, with a few exceptions. The fluid routing plug **950** comprises a body **962** having a first outer surface **964** joined to a second outer surface **952** by an intermediate outer surface **966**. In contrast to the fluid routing plug **900**, the second surface **952** of the fluid routing plug **950** is formed identically to the second surface **856** of the fluid routing plug **850**, shown in FIGS. **100A-100E**. A central base **954** formed in the second surface **952** is spaced from an edge **956** of a tapered wall **958** such that a throat **960** is formed within the second surface **952**. The throat **960** serves the same purpose as the throat **566** formed in the fluid routing plug **550**.

(492) Continuing with FIGS. **128A-128G**, a plurality of first fluid passages **968**, shown in FIG. **128G**, and a plurality of second fluid passages **970**, shown in FIG. **128A**, are formed in the body **962**. The first and second fluid passages **968** and **970** are identical to the first and second passages **910** and **924** formed in the fluid routing plug **900**. However, as discussed above, an opening **972** of the first fluid passages **968** may extend along a non-zero angle relative to a central longitudinal axis **974** of the body **962**, as shown in FIG. **128D**. In alternative embodiments, the openings **972** may be identical to the openings **918**, shown in FIG. **121**. Like the fluid routing plug **900**, the angle at which the first fluid passages **968** are formed in the body **962** may vary, as desired, in order to increase the wall thickness within the body **962** and reduce stress in the body **962** during operation.

(493) In alternative embodiments, the first and second surfaces **964** and **952** of the fluid routing plug **950** may be configured like the fluid routing plug **600**. In further alternative embodiments, the first and second surfaces of the fluid routing plug **950** may be identical to the first and second surfaces of the fluid routing plug **116**.

(494) Turning to FIGS. **129-131**, another embodiment of a fluid routing plug **1000** is shown. The fluid routing plug **1000** may be installed within the housing **104** in place of the fluid routing plug **116**. The fluid routing plug **1000** is identical to the fluid routing plug **116**, but the fluid routing plug **1000** includes a first and second annular recess **1002** and **1004** formed in its intermediate surface **1001**. The first annular recess **1002** is positioned between a first bevel **1006** and a first sealing surface **1008**. The second annular recess **1004** is positioned between a second sealing surface **1010** and a second bevel **1012**.

(495) When the fluid routing plug **1000** is installed within the horizontal bore **106**, a small annular space exists between the wall of the housing **104** and each recess **1002** and **1004**. The space provides relief areas for excess fluid or proppant to collect during operation. The first and second recesses **1002** and **1004** may also be formed in the intermediate surfaces of the fluid routing plugs **550**, **600**, **700**, **800**, **900**, and **950**.

(496) In alternative embodiments, the first and second surfaces of each of the fluid routing plugs **550**, **600**, **700**, **800**, **900**, and **950** may each be sized so as to engage with identically sized suction and discharge valves **292**, **294**, **614** or **616**, as discussed with regard to fluid routing plug **600**. In further alternative embodiments, the first and second surfaces of each of the fluid routing plugs **550**, **600**, **700**, **800**, **900**, and **950** may be sized so as to engage with differently sized suction and discharge valves **292**, **294**, **614** or **616**. In such embodiment, the valves **292**, **294**, **614** or **616** may be sized as desired, as long as the ratio of the outer sealing diameter A to the inner sealing diameter B of each valve is preferably 1.55 or greater, as discussed with regard to FIGS. **72** and **85**. The desired size of the valve may vary depending on the desired fluid velocity within the corresponding fluid routing plug.

(497) With reference to FIGS. **132-138**, another embodiment of a fluid routing plug **1100** is shown. The fluid routing plug **1100** is identical to the fluid routing plug **116**, with the exception of its outer intermediate surface **1102**. The intermediate surface **1102** includes a first sealing surface **1104**, a second sealing surface **1106**, and a first bevel **1108**. In contrast to the fluid routing plug **116**, the intermediate surface **1102** does not include a second bevel. The first bevel **1108** is positioned between the first sealing surface **1104** and a first surface **1110** of the fluid routing plug **1100**. In contrast, the first bevel **378** is positioned between the first fluid passage **326** and the first sealing surface **370** in the fluid routing plug **116**, as shown in FIG. **64**.

(498) A plurality of first and second fluid passages **1112** and **1114** are formed in the fluid routing plug **1100**. The first and second fluid passages **1112** and **1114** are formed identically to the first and second fluid passages **326** and **336** formed in the fluid routing plug **116**. Likewise, the first surface **1110** and a second surface **1109** of the fluid routing plug **1100** are generally identical to the first and second surfaces **318** and **320** of the fluid routing plug **116**. The intermediate surface **1102** also includes a taper **1116** between the second sealing surface **1104** and a plurality of openings **1118** of the first fluid passages **1112**. Fluid is routed throughout the fluid routing plug **1100** and the housing

1120 in the same manner as the fluid routing plug **116** and housing **104** shown in FIGS. **50** and **51**. (499) Continuing with FIGS. **136-138**, another embodiment of a housing **1120** is shown. The housing **1120** is identical to the housing **104** with the exception of the shape of the walls surrounding its horizontal bore **1122**. The walls surrounding the horizontal bore **1122** are shaped to mate with the fluid routing plug **1100**. The walls include a first beveled surface **1124** that mates with the first bevel **1108**, as shown in FIG. **138**. Unlike the housing **104**, the walls do not include a second beveled surface, as shown in FIG. **137**. When the fluid routing plug **1100** is installed within the housing **1120**, an annular chamber **1126** is formed between the walls of the housing **1120** and the taper **1116** on the intermediate surface **1102**. The chamber **1126** provides a pathway for fluid to flow from a pair of intake bores **1128** and **1130** and the openings **1118**. A first and second seal **1132** and **1134** installed within the housing **1120** are identical to the first and second seals **374** and **376**, shown in FIGS. **70** and **71**.

(500) With reference to FIGS. **139-142**, another embodiment of a fluid routing plug **1200** is shown. The fluid routing plug **1200** is identical to the fluid routing plug **1100**, with the exception of its first and second surfaces **1202** and **1204**. The first surface **1202** does not include an outer rim. Rather, the first surface **1202** has a tapered wall **1206** that tapers between an entrance **1208** of an axially blind bore **1210** and an outer edge **1212** of the first surface **1202**. A plurality of openings **1214** for a plurality of second fluid passages **1216** are formed in the tapered wall **1206** adjacent the outer edge **1212**. Removing the outer rim provides more surface area for the tapered wall **1206** and allows the tapered wall **1206** to taper at an angle of up to 60 degrees. Providing more surface area for the tapered wall **1206** provides more sealing area for a suction valve.

(501) Similarly, the second surface **1204** does not include an outer rim. Rather, the second surface **1204** has a tapered wall **1218** that tapers between a central base **1220** and an outer edge **1222** of the second surface **1204**. Removing the outer rim provides more surface area for the tapered wall **1218** and allows the tapered wall **1218** to taper at an angle of up to 60 degrees. Providing more surface area for the tapered wall **1218** provides more sealing area for a discharge valve. The fluid routing plug **1200** may be installed within the housing **1120**, shown in FIG. **136**. Fluid is routed throughout the fluid routing plug **1200** and the housing **1120** in the same manner as the fluid routing plug **116** and housing **104**, shown in FIGS. **50** and **51**.

(502) With reference to FIGS. **143-150**, another embodiment of a fluid routing plug **1300** is shown. The fluid routing plug **1300** comprises a body **1302** having opposed first and second outer surfaces **1304** and **1306** joined by an intermediate outer surface **1308**. A central longitudinal axis **1310** extends through the body **1302** and both surfaces **1304** and **1306**. A plurality of first fluid passages **1312** are formed within the body **1302** and interconnect the intermediate surface **1308** and the first surface **1304**, as shown in FIG. **146**. The first fluid passages **1312** interconnect the intermediate surface **1308** and the first surface **1304** by way of an axially blind bore **1314**, as shown in FIG. **146**. The first fluid passages **1312** have an oval cross-sectional shape similar to the first fluid passages **326**, shown in FIG. **59**. In contrast to the first fluid passages **326**, the first fluid passages **1312** extend orthogonally to the central longitudinal axis **1310** of the body **1302**.

(503) Continuing with FIG. **147**, a plurality of second fluid passages **1316** are formed in the body **1302** that interconnect the first surface **1304** and the second surface **1306**. Like the second fluid passages **336**, shown in FIGS. **60-63**, the second fluid passages **1316** extend along a straight-line path between the first and second surfaces **1304** and **1306**. The second fluid passages **1316** are offset from and do not intersect the first fluid passages **1312**.

(504) Continuing with FIGS. **147** and **149**, a first recessed cavity **1318** is formed in the first surface **1304**. The first cavity **1318** is characterized by a base **1320** surrounded by a side wall **1322**. The base **1320** extends between the side wall **1322** and an entrance **1324** of the blind bore **1314**. The side wall **1322** extends between the base **1320** and an outer rim **1326** of the first surface **1304**. A plurality of openings **1328** for the second fluid passages **1316** are formed in the outer rim **1326**. The openings **1328** are positioned in groups **1330**, similar to the openings **348**, shown in FIG. **56**.

(505) An annular first insert **1332** is installed within the first cavity **1318** such that it is fittingly received within the first cavity **1318**. The first insert **1332** has a base **1334** joined to a sidewall **1336** and a central opening **1338**. The base **1334** and sidewall **1336** are sized to correspond with the base **1320** and sidewall **1322** of the first recessed cavity **1318**. The first insert **1332** has a tapered surface **1340** that extends between the sidewall **1336** and the base **1334**. The tapered surface **1340** is sized to engage a suction valve **1342**, as shown in FIG. 150.

(506) The first insert **1332** is made of a harder and more wear-resistant material than the body **1302**. For example, if the body **1302** is made of metal, the first insert **1332** may be made of tungsten carbide. Making the first insert **1332** out of a harder and more wear-resistant material than the body **1302** helps extend the life of the body **1302**. Further, the first insert **1332** may be removed and replaced with a new insert, if needed.

(507) Continuing with FIGS. 145, 147, and 148, a central base **1344** is formed in the second surface **1306** of the body **1302**. A blind bore **1348** is formed in the central base **1344**, similar to the blind bore **362** shown in FIG. 55. A plurality of openings **1350** for the second fluid passages **1316** are formed in the central base **1344** and surround the blind bore **1348**. In contrast, to the openings **364**, shown in FIG. 54, the openings **1350** are spaced around in the blind bore **1348** in groups **1352**, similar to the openings **1328** on the first surface **1304**.

(508) A second recessed cavity **1354** is formed in the second surface **1306** of the body **1302** and surrounds the openings **1350**. The second recessed cavity **1354** is characterized by a base **1356** surrounded by a sidewall **1358**. The base **1356** of the second cavity **1354** serves as an extension of the central base **1344**. The sidewall **1358** joins the base **1356** and an outer rim **1360**. An annular second insert **1362** is installed within the second cavity **1354** such that the second insert is fittingly received within the second cavity **1354**. The second insert **1362** is identical to the first insert **1332**, but it has an outer diameter that is greater than an outer diameter of the first insert **1332**. Likewise, a central passage **1331** formed in the second insert **1362** has a greater diameter than the central passage **1338**. A tapered surface **1364** of the second insert **1362** is configured to engage a discharge valve **1366**, as shown in FIG. 150.

(509) Continuing with FIGS. 143, 144, and 146, the intermediate surface **1308** of the fluid routing plug **1300** has a generally cylindrical shape. The intermediate surface **1308** includes a first sealing surface **1368** and a second sealing surface **1370**. The sealing surfaces **1368** and **1370** are configured to engage respective first and second seals **1372** and **1374** installed within another embodiment of a housing **1376**, as shown in FIG. 150. The seals **1372** and **1374** may be identical to the first and second seals **374** and **376** installed within the housing **104**. The first sealing surface **1368** has a smaller diameter than the second sealing surface **1370**. The intermediate surface **1308** also includes a first bevel **1378** positioned between the first sealing surface **1368** and the first surface **1304**, and a second bevel **1380** positioned between the second sealing surface **1374** and the first fluid passages **1312**.

(510) With reference to FIG. 150, the housing **1376** is similar to the housing **104**, with the exception of the shape of its horizontal bore **1382**. The walls surrounding the horizontal bore **1382** are shaped to engage the fluid routing plug **1300**. The walls engage the outer rims **1326** and **1360** on the first and second surfaces **1304** and **1306** and the first and second bevels **1378** and **1380**. Fluid is routed throughout the fluid routing plug **1300** and the housing **1376** in the same manner as the fluid routing plug **116** and housing **104**, shown in FIGS. 50 and 51.

(511) With reference to FIGS. 151-160, another embodiment of a fluid routing plug **1400** is shown. The fluid routing plug **1400** comprises a body **1402** having opposed first and second outer surfaces **1404** and **1406** joined by an intermediate outer surface **1408**. A central longitudinal axis **1410** extends through the body **1402** and both surfaces **1404** and **1406**, as shown in FIG. 155. A plurality of first fluid passages **1412** are formed within the body **1402** and interconnect the intermediate surface **1408** and the first surface **1404**, as shown in FIG. 156. The first fluid passages **1412** interconnect the intermediate surface **1408** and the first surface **1404** by way of a first axially blind

bore **1414**, as shown in FIG. **156**. The first blind bore **1414** opens on the first surface **1404** of the body **1402**.

(512) Rather than having a plurality of wide first fluid passages, like those shown in FIG. **55**, the first fluid passages **1412** comprise a plurality of groups **1416** of smaller first fluid passages **1412**. The first fluid passages **1412** within each group **1416** are positioned in a side-by-side relationship, as shown by openings **1417** for the first fluid passages **1412** on the intermediate surface **1408** in FIG. **164**. Three first fluid passages **1412** are shown in each group **1416** in FIGS. **151-154**. In alternative embodiments, more or less than three first fluid passages may be included in each group. The first fluid passages **1412** each have a circular cross-sectional shape and extend orthogonally to the central longitudinal axis **1410** of the body **1402**, as shown in FIG. **156**.

(513) The fluid routing plug **1400** further comprises a plurality of second fluid passages **1418** formed within the body **1402**, as shown in FIG. **155**. The second fluid passages **1418** interconnect the first surface **1404** and the second surface **1406** of the body **1402**. The second fluid passages **1418** interconnect the first surface **1404** and the second surface **1406** by way of a second axially blind bore **1420**. The second blind bore **1420** opens on the second surface **1406** of the body **1402**. Each second fluid passage **1418** extends between the first surface **1404** and the second blind bore **1420** along a straight line path.

(514) Continuing with FIG. **158**, a first recessed cavity **1422** is formed in the first surface **1404** of the body **1402**. The first cavity **1422** is identical to the first cavity **1318** shown in FIG. **149**. An annular first insert **1424** is installed within the first cavity **1422**. The first insert **1424** is identical to the first insert **1332**, shown in FIG. **149**. The first insert **1424** is configured to engage a suction valve **1426**, as shown in FIGS. **159** and **160**.

(515) The first cavity **1422** is surrounded by an outer rim **1428**. A plurality of openings **1430** for the second fluid passages **1418** are formed on the outer rim **1428**. The openings **1430** are equally spaced around the outer rim **1428**.

(516) Continuing with FIG. **157**, a second recessed cavity **1438** is formed in the second surface **1406** and surrounds an entrance **1440** to the second blind bore **1420**. The second cavity **1438** is identical to the second cavity **1354**, shown in FIG. **148**. An annular second insert **1442** is installed within the second cavity **1438**. The second insert **1442** is identical to the second insert **1362**, shown in FIG. **148**. The second insert **1442** is configured to engage a discharge valve **1444**, as shown in FIGS. **159** and **160**.

(517) Continuing with FIG. **155**, a third blind bore **1434** is formed in a base **1436** of the second blind bore **1420**. The third blind bore **1434** is configured to engage with a tool to help remove the fluid routing plug **1400** from the horizontal bore with which it is installed.

(518) Turning back to FIGS. **151**, **152**, and **154**, the intermediate surface **1408** of the body **1402** varies in diameter throughout its length and has a generally cylindrical shape. The intermediate surface **1408** includes a first sealing surface **1446** and a second sealing surface **1448**. The first and second sealing surfaces **1446** and **1448** are configured to engage with a first and second seal **1450** and **1452** installed within another embodiment of a housing **1454**, as shown in FIGS. **159** and **160**. The first and second seal **1450** and **1452** may be identical to the first and second seals **374** and **376** installed within the housing **104**, as shown in FIGS. **70** and **71**. The first sealing surface **1446** is positioned between the first surface **1404** of the body **1402** and the first fluid passages **1412**. The second sealing surface **1448** is positioned between the second surface **1406** of the body **1402** and the first fluid passages **1412**.

(519) The intermediate surface **1408** has a first reduced diameter portion **1456** around the first fluid passages **1412**, as shown in FIG. **154**. When the fluid routing plug **1400** is installed within the housing **1454**, an annular chamber **1458** is formed between the first reduced diameter portion **1456** and the housing **1454**, as shown in FIG. **159**. An intake bore **1460** opens into the chamber **1458**. The chamber **1458** provides a pathway for fluid to flow from the intake bore **1460** to the openings **1417** of the first fluid passages **1412**.

(520) The intermediate surface **1408** also has a second reduced diameter portion **1464**, as shown in FIG. **154**. The second reduced diameter portion **1464** surrounds the second cavity **1438**, as shown in FIG. **156**. The second reduced diameter portion **1464** provides space for a plurality of legs **1466** of a discharge valve guide **1468** to engage the fluid routing plug **1400**, as shown in FIG. **159**. The legs **1466** help hold the components installed within the housing **1454** in place during operation. (521) With reference to FIGS. **159** and **160**, the housing **1454** is generally similar to the housing **104** with the exception of the shape of its horizontal bore **1470**. While not shown, a stuffing box and related components may be attached to the housing **1454**. The walls surrounding the horizontal bore **1470** are shaped to engage the fluid routing plug **1400**. The first surface **1404** of the fluid routing plug **1400** seats against an annular shoulder **1472** formed in the housing **1454**. Fluid is routed throughout the fluid routing plug **1400** and the housing **1454** in the same manner as the fluid routing plug **116** and housing **104** shown in FIGS. **50** and **51**. A discharge bore **1474** is shown opening on a bottom surface **1476** of the housing **1454** in FIGS. **159** and **160**. In alternative embodiments, the discharge bore may open on a top surface of the housing, like the discharge bore **178** shown in FIGS. **50** and **51**. The housing **1454** shown in FIGS. **159** and **160** also includes a single intake bore **1460**. In alternative embodiments, a second intake bore may also be formed in the housing, like the housing **104**.

(522) With reference to FIGS. **161-164**, another embodiment of a fluid routing plug **1500** is shown. The fluid routing plug **1500** is identical to the fluid routing plug **1400**, but it does not include a first and second sealing surface. Instead, first and second annular grooves **1502** and **1504** are formed in an intermediate surface **1506**. The grooves **1502** and **1504** are formed at the same position as the first and second sealing surface **1446** and **1448** formed in the fluid routing plug **1400**, as shown in FIG. **159**. First and second seals **1508** and **1510** are installed within each respective first and second groove **1502** and **1504**.

(523) The fluid routing plug **1500** is installed within another embodiment of a housing **1512**, as shown in FIG. **164**. The housing **1512** is identical to the housing **1454**, but it does not include any grooves for housing the first and second seals **1508** and **1510**. Rather, the first and second seals **1508** and **1510** engage the walls of the housing **1512** when the fluid routing plug **1500** is installed. Thus, the seals **1508** and **1510** are installed within the fluid routing plug **1500**, rather than the walls of the housing **1512**.

(524) Continuing with FIGS. **161-163**, the fluid routing plug **1500** also does not include the second reduced diameter portion formed in the fluid routing plug **1400**, as shown in FIG. **154**. Instead, the fluid routing plug **1500** has a uniform diameter between the second seal **1510** and its second surface **1514**. A discharge valve guide **1516** may engage the second surface **1514** of the fluid routing plug **1500**, as shown in FIG. **164**.

(525) With reference to FIGS. **165-170**, another embodiment of a fluid routing plug **1600** is shown. The fluid routing plug **1600** comprises a body **1602** having opposed first and second outer surfaces **1604** and **1606** joined by an intermediate outer surface **1608**. A central longitudinal axis **1610** extends through the body **1602** and both surfaces **1604** and **1606**, as shown in FIG. **168**. A plurality of first fluid passages **1612** are formed within the body **1602** and interconnect the intermediate surface **1608** and the first surface **1604**. The first fluid passages **1612** extend at a non-zero angle relative to the central longitudinal axis **1610** of the body **1602**.

(526) Continuing with FIGS. **168** and **169**, a first recessed cavity **1614** is formed in the first surface **1604**. The first cavity **1614** is identical to the first cavity **1318**, shown in FIG. **149**, but includes a plurality of openings **1616** for the first fluid passages **1612**, as shown in FIG. **169**. An annular first insert **1618** is installed within the first cavity **1614**. The first insert **1618** is identical to the first insert **1332**, but it includes a plurality of insert passages **1620**. When the first insert **1618** is installed within the first cavity **1614**, the plurality of insert passages **1620** align with the plurality of first fluid passages **1612** in a one-to-one relationship, as shown in FIG. **168**. Thus, the insert passages **1620** become an extension of the first fluid passages **1612**.

(527) An opening **1622** of each first fluid passage **1612** on the intermediate surface **1608** has an oval shape, while an opening **1624** of each first fluid passage **1612** on the first surface **1604** has a circular shape, as shown in FIGS. **165** and **166**. The openings **1624** are equally spaced around the periphery of the first insert **1618**, as shown in FIG. **167**. The fluid routing plug **1600** includes more first fluid passages **1612** than the fluid routing plug **116** shown in FIG. **52**. However, the first fluid passages **1612** have a smaller diameter than a diameter of the first fluid passages **326** formed in the fluid routing plug **116**.

(528) Continuing with FIG. **168**, the fluid routing plug **1600** further comprise a single second fluid passage **1626** formed in the body **1602**. The second fluid passage **1626** interconnects the first surface **1604** and the second surface **1606** along a straight-line path. The second fluid passage **1626** has a circular cross-sectional shape and extends along the central longitudinal axis **1610** of the body **1602**. Because the body **1602** only includes a single second fluid passage **1626**, the second fluid passage **1626** has a much larger diameter than the other second fluid passages described herein.

(529) The second surface **1606** of the body **1602** includes a second recessed cavity **1628** having an annular second insert **1630** installed therein. The second cavity **1628** and second insert **1630** are identical to the second cavity **1354** and second insert **1362**, shown in FIG. **148**. The first cavity **1614** and the first insert **1618** surround an opening **1632** of the second fluid passage **1626** on the first surface **1604** of the body **1603**. Likewise, the second cavity **1628** and the second insert **1630** surround an opening **1634** on the second surface **1606** of the body **1602**.

(530) The intermediate surface **1608** of the fluid routing plug **1600** is identical to the intermediate surface **1506** of the fluid routing plug **1500**, but it does not have seals installed within grooves formed in the intermediate surface **1608**. Rather, the intermediate surface **1608** has a first and second sealing surface **1636** and **1638**, like the fluid routing plug **1400**. In alternative embodiments, a first and second seal may be installed within the intermediate surface **1608** of the fluid routing plug **1600**.

(531) With reference to FIG. **170**, fluid is routed through the fluid routing plug **1600** differently from the previously described fluid routing plugs. The fluid routing plug **1600** is shown installed within the housing **1454**. Fluid passes from the intake bore **1460**, through the first fluid passages **1612** and into an opening in the horizontal bore **1470**. A suction valve **1640** used with the fluid routing plug **1600** includes a central passage **1642**. Thus, the suction valve **1642** can seal the first fluid passages **1612** while leaving the second fluid passage **1626** open to the horizontal bore **1470**. As a plunger extends, the plunger forces fluid in the horizontal bore **1470** into the second fluid passage **1626**. The pressurized fluid flows through the second fluid passage **1626** and exits the second surface **1606** of the body **1602**. From there, the pressurized fluid flows around a discharge valve **1644** and towards the discharge bore **1474** formed in the housing **1454**.

(532) Various features of the fluid routing plugs **116**, **600**, **700**, **800**, **900**, **1000**, **1100**, **1200**, **1400**, **1500**, and **1600** described above may be modified, varied, or included in each of the plugs, as desired.

(533) Turning to FIGS. **171-187**, another embodiment of a fluid routing plug **1700** is shown. Instead of having a single body, like the fluid routing plugs described above, the fluid routing plug **1700** is separated into two different sections—a first section **1702** and a second section **1704**. A suction valve **1706** seals against the first section **1702** and a discharge valve **1708** seals against the second section **1704**, as shown in FIGS. **185** and **187**. The first section **1702** may be characterized as a suction seat, and the second section **1704** may be characterized as a discharge seat. In operation, fluid is primarily routed throughout the first section **1702**.

(534) Continuing with FIGS. **171-180**, the first section **1702** comprises a body **1710** having opposed first and second surfaces **1712** and **1714** joined by an intermediate surface **1716**. A plurality of first fluid passages **1718** are formed in the body **1710** and interconnect the intermediate surface **1716** and the first surface **1712**, as shown in FIG. **173**. The first fluid passages **1718**

interconnect the intermediate surface **1716** and the first surface **1712** by way of an axially blind bore **1720**. The first fluid passages **1718** have the same shape and orientation as the first fluid passages **1312**, shown in FIG. **146**.

(535) Continuing with FIG. **174**, a plurality of second fluid passages **1722** are formed in the body **1710** and interconnect the first and second surfaces **1712** and **1714** along a straight-line path. The first fluid passages **1718** and the second fluid passages **1722** are formed offset from one another in the body **1710** and do not intersect. The second fluid passages **1722** each have a circular cross-sectional shape, like the second fluid passages **336**, shown in FIGS. **60-63**. The second fluid passages **1722** may each extend at a non-zero angle relative to a central longitudinal axis **1724** of the body **1710**.

(536) Continuing with FIGS. **179** and **180**, a first recessed cavity **1726** is formed in the first surface **1712** of the first section **1702**. The first cavity **1726** is characterized by a base **1728** surrounded by a sidewall **1730**. The sidewall **1730** is joined to an outer rim **1732**. A first annular insert **1734** is installed within the first cavity **1726**, as shown in FIG. **179**. The first insert **1734** is identical to the first insert **1332**, shown in FIG. **149**. The first insert **1734** is configured to engage the suction valve **1706**, as shown in FIGS. **185** and **187**.

(537) Continuing with FIGS. **175** and **177**, each second fluid passage **1722** opens on the outer rim **1732** of the first surface **1712**, as shown by a plurality of openings **1736**. The openings **1736** may be spaced in groups **1738** on the first surface **1712**, similar to the openings **1328**, shown in FIG. **144**. Each group **1738** shown in FIG. **177** includes three openings **1736**. A plurality of blind bores **1740** of varying shapes and sizes are formed in the outer rim **1732** between each of the openings **1736**. The blind bores **1740** function to remove excess weight from the first section **1702**.

(538) Continuing with FIGS. **171** and **172**, the second surface **1714** of the first section **1702** has an outer rim **1742** and a central blind bore **1744**. The blind bore **1744** may be threaded and is configured to engage with a tool used to grip the first section **1702**. A plurality of countersinks **1746** are formed in the second surface **1714** surrounding the blind bore **1744**. The countersinks **1746** form a plurality of raised surfaces **1748** surrounding the blind bore **1744**. Each raised surface **1748** may have a generally triangular shape. The countersinks **1746** help remove excess weight from the first section **1702**. In alternative embodiments, the countersinks and raised areas may have various shapes and sizes as desired.

(539) The second fluid passages **1722** open on the raised surfaces **1748**, as shown by a plurality of openings **1750**. A plurality of blind bores **1752** of various shapes and sizes may be formed in the raised surfaces **1748** around the openings **1750**. The blind bores **1752** help remove excess weight from the first section **1702**. The openings **1750** may remain spaced in groups **1738** on the second surface **1714**. In alternative embodiments, the openings of the second fluid passages may have different spacing or grouping, as desired.

(540) Continuing with FIGS. **176** and **178**, the intermediate surface **1716** of the first section **1702** has a generally cylindrical shape. The intermediate surface **1716** includes a first sealing surface **1754** positioned between the first surface **1712** and the first fluid passages **1718**, and a second sealing surface **1756** positioned between the second surface **1714** and the first fluid passages **1718**. The first and second sealing surfaces **1754** and **1756** are configured to engage a first and second seal **1758** and **1760** installed within another embodiment of a housing **1762**, as shown in FIGS. **185** and **187**. The first and second seals **1758** and **1760** may be identical to the first and second seals **374** and **376** installed within the housing **104**, as shown in FIGS. **70** and **71**. The first sealing surface **1754** has a smaller diameter than the second sealing surface **1756** in order to reduce friction during installation of the first section **1702** within the housing **1762**.

(541) The intermediate surface **1716** further includes a first bevel **1764** positioned between the first sealing surface **1754** and the first surface **1712**. The intermediate surface **1716** further includes a taper **1766** positioned between the second sealing surface **1756** and the first fluid passages **1718**. The first bevel **1764** is configured to engage a first beveled surface **1768** formed in the housing

1762, as shown in FIG. **187**. When the first section **1702** is installed within the housing **1762**, an annular chamber **1770** is formed between the taper **1766** and the housing **1762**, as shown in FIG. **187**. The chamber **1770** provides a pathway for fluid from an upper and lower intake bore **1772** and **1774** and the first fluid passages **1718**.

(542) Continuing with FIGS. **185** and **187**, fluid is routed throughout the first section **1702** in the same manner as the fluid routing plug **116**. However, when fluid exits the second surface **1714** of the first section **1702**, the fluid is directed into the second section **1704**, instead of being directed towards a discharge bore.

(543) Turning to FIGS. **181-184**, the second section **1704** comprises a body **1776** having opposed first and second surfaces **1778** and **1780** joined by an intermediate surface **1782**. A central passage **1784** interconnects the first and second surfaces **1778** and **1780** of the body **1776**. The first surface **1778** has an outer rim **1786** joined to a tapered wall **1788**. The tapered wall **1788** extends between the central passage **1784** and the outer rim **1786**. A second recessed cavity **1790** is formed in the second surface **1780**. The second cavity **1790** is characterized by a base **1792** surrounded by a sidewall **1794**. The sidewall **1794** is joined to an outer rim **1796**. The outer rim **1796** is thicker than the outer rim **1786**. The base **1792** of the second cavity **1790** surrounds an opening **1798** of the central passage **1784**.

(544) A second annular insert **1800** is installed within the second cavity **1790** and surrounds the opening **1798** of the central passage **1784**, as shown in FIGS. **185** and **187**. The second insert **1800** is identical to the second insert **1362**, shown in FIG. **148**. The second insert **1800** is configured to engage the discharge valve **1708**, as shown in FIGS. **185** and **187**.

(545) Continuing with FIGS. **181-184**, the intermediate surface **1782** of the second section **1704** has a generally cylindrical shape. The intermediate surface **1782** includes a third sealing surface **1802** positioned between the first and second surfaces **1778** and **1780**, as shown in FIG. **184**. The third sealing surface **1802** is configured to engage with a third seal **1804** installed within the housing **1762**, as shown in FIG. **187**. The third seal **1804** may be identical to the first or second seal **1758** or **1760**. The intermediate surface **1782** also includes a second bevel **1806** positioned between the third sealing surface **1802** and the second surface **1780**. The second bevel **1806** is configured to engage with a second beveled surface **1808** formed in the housing **1762**, as shown in FIG. **187**.

(546) Continuing with FIGS. **185-187**, the second section **1704** is installed within the housing **1762** such that the outer rim **1786** of its first surface **1778** engages the outer rim **1742** of the second surface **1714** of the first section **1702**. A seal **1810** may be positioned between the outer rims **1786** and **1742**, as shown in FIG. **186**. The seal **1810** may be installed within a groove **1812** formed in the outer rim **1786** of the second section **1704**. The first section **1702** may have smaller outer diameters than the second section **1704** in order to reduce friction as the first and second sections **1702** and **1704** are installed within the housing **1762**, as shown in FIG. **187**.

(547) In operation, fluid exiting the second surface **1714** of the first section **1702** is funneled into the central passage **1784** of the second section **1704** by way of the tapered wall **1788**. Fluid exits the second surface **1780** of the second section **1704** and flows around the discharge valve **1708** and a discharge valve guide **1814** towards a discharge bore **1816**.

(548) With reference to FIGS. **188-195**, another embodiment of a first section **1900** is shown. The first section **1900** is identical to the first section **1702** with a few exceptions. A plurality of first fluid passages **1902** formed in a body **1904** of the first section **1900** have an oblong cross-sectional shape, rather than an oval cross-sectional shape, as shown by the opening **1903** in FIG. **194**. An end of the oblong shape may be wider than its opposed end, as shown in FIG. **194**.

(549) A first surface **1906** of the first section **1900** is identical to the first surface **1712** of the first section **1702** shown in FIG. **177**, with the exception of its blind bores **1908**. The blind bores **1908** have different spacing than the blind bores **1740** formed in the first surface **1712**. A second surface **1910** of the first section **1900** includes a central base **1912** surrounded by an outer rim **1914**. A blind bore **1916** is formed in the central base **1912**, similar to the blind bore **1744**.

(550) Instead of including a plurality of countersinks and raised surfaces like the first section **1702**, the second surface **1910** of the first section **1900** includes a cut-out area **1918** between the blind bore **1916** and the outer rim **1914**. The cut-out area **1918** helps remove excess weight from the first section **1900**. A plurality of second passages **1920** formed in the first section **1900** open on the outer rim **1914** of the second surface **1910**, as shown by the openings **1922**. A plurality of blind bores **1924** are positioned adjacent the openings **1922** to help remove excess weight. The spacing of the openings **1922** may be identical to the spacing of the openings **1750**, shown in FIG. **172**.

(551) An intermediate surface **1926** of the first section **1900** is identical to the intermediate surface **1716** of the first section **1702** with a minor exception. The intermediate surface **1926** includes two tapers **1928** and **1930** positioned on opposite sides of the first fluid passages **1902**. When the first section **1900** is installed in the housing **1762**, an annular chamber **1932** is formed between the housing **1762** and the tapers **1928** and **1930**, as shown in FIG. **195**. The annular chamber **1932** provides a pathway for fluid from the intake bores **1772** and **1774** to the first fluid passages **1902**.

(552) Continuing with FIG. **195**, the second section **1704** is shown engaging the first section **1900** within another embodiment of a housing **1934**. The housing **1934** is generally identical to the housing **1762** shown in FIG. **187**. A discharge conduit **1936** is shown attached to a bottom surface **1938** of the housing **1934**. In alternative embodiments, the discharge conduit may attach to a top surface of the housing. Likewise, the discharge bore may interconnect the top surface of the housing and the horizontal bore. While not shown, a stuffing box and corresponding components may be attached to the housing **1934**. Fluid is routed through the first and second sections **1900** and **1704** in the same manner as the fluid routing plug **1700**.

(553) With reference to FIGS. **196-203**, another embodiment of a fluid routing plug **2000** is shown. The fluid routing plug **2000** comprises a first section **2002** and a second section **2004**. The first section **2002** is generally identical to the first section **1900**, with a few exceptions. In contrast to the first surface **1900**, no blind bores are formed in an outer rim **2006** of a first surface **2008** of the first section **2002**. An opening **2010** of each of the plurality of second fluid passages **2012** formed in the first section **2002** are equally spaced around the outer rim **2006**, rather than being spaced in groups. (554) A second surface **2014** of the first section **2002** is flat with the exception of a plurality of openings **2016** of the second fluid passages **2012** and a central blind bore **2018**. The openings **2016** are equally spaced around the blind bore **2018**. The central blind bore **2018** may be threaded and configured to engage a tool used to grip the first section **2002**. A plurality of first fluid passages **2020** formed in the first section **2002** are identical to the first fluid passages **1902**, shown in FIG. **190**. However, an oblong shaped opening **2021** of each first fluid passage **2020** may have the same sized ends, as shown in FIG. **199**.

(555) Turning to FIGS. **202** and **203**, the second section **2004** is shown engaging the first section **2002** within another embodiment of a housing **2022**. The second section **2004** is generally identical to the second section **1704**, shown in FIGS. **181-184**, but its first surface **2024** may have a thicker outer rim **2026** than the second section **1704**. While not shown in FIGS. **202** and **203**, a seal may be placed between the first and second sections **2002** and **2004**, like that shown in FIG. **186**. The housing **2022** is sized to receive the first and second sections **2002** and **2004**. The housing **2022** shown in FIGS. **202** and **203** has a single intake bore **2028** and a discharge bore **2030** positioned on a bottom surface **2032** of the housing **2022**. In alternative embodiments, the housing may include a second intake bore and the discharge bore may be formed on a top surface of the housing. While not shown, a stuffing box and corresponding components may be attached to the housing **2022**. Fluid is routed throughout the first and second sections **2002** and **2004** in the same manner as the fluid routing plug **1700**, shown in FIGS. **185** and **187**.

(556) With reference to FIGS. **204-211**, another embodiment of a first section **2100** is shown. The first section **2100** is identical with the first section **2002**, with a few minor exceptions. An intermediate surface **2102** of the first section **2100** has a uniform diameter throughout its length. The intermediate surface **2102** includes a first sealing surface **2104** positioned adjacent its first

surface **2106** and a second sealing surface **2108** positioned adjacent its second surface **2110**. The first and second sealing surfaces **2104** and **2108** are configured to engage a first and second seal **2112** and **2114** installed within another embodiment of a housing **2116**, shown in FIGS. **210** and **211**. The first and second seal **2112** and **2114** may be identical to the first and second seals **374** and **376**, shown in FIGS. **70** and **71**.

(557) A plurality of first fluid passages **2118** formed in the first section **2100** may be generally identical to the first fluid passages **1902**, shown in FIG. **190**. The second surface **2110** of the first section **2100** shown in FIG. **205** does not include a central blind bore. In alternative embodiments, a central blind bore may be formed in the second surface. A plurality of second fluid passages **2120** formed in the first section **2100** may extend parallel to a central longitudinal axis **2122** of the first section **2100**, as shown in FIG. **208**.

(558) Turning to FIGS. **210** and **211**, the second section **2004** is shown engaging the first section **2100** within the housing **2116**. While not shown, a seal may be placed between the first and second sections **2100** and **2004**, like that shown in FIG. **186**. The housing **2116** is sized to receive the first and second sections **2100** and **2004**. The housing **2116** shown in FIGS. **210** and **211** has a single intake bore **2124** and a discharge bore **2126** positioned on a bottom surface **2128** of the housing **2116**. In alternative embodiments, the housing may include a second intake bore and the discharge bore may be formed on a top surface of the housing. While not shown, a stuffing box and corresponding components may be attached to the housing **2116**. Fluid is routed throughout the first and second sections **2100** and **2004** in the same manner as the fluid routing plug **1700**, shown in FIGS. **185** and **187**.

(559) With reference to FIGS. **212-222**, another embodiment of a fluid routing plug **2500** is shown. The fluid routing plug **2500** is generally identical to the fluid routing plug **550**, shown in FIG. **100A-100E**, but the plug **2500** includes modified first and second surfaces **2502** and **2504**. The first surface **2502** comprises a recessed area **2505** positioned between the opening of an axially blind bore **2506** and a tapered surface **2508**. The recessed area **2505** is configured to receive a hardened insert **2510**, shown in FIG. **220**.

(560) Similarly, the second surface **2504** comprises a recessed area **2512** positioned between a plurality of openings **2514** of a plurality of second fluid passages **2516** and a tapered surface **2518**. The recessed area **2512** is configured to receive a hardened insert **2520**. The inserts **2510** and **2520** are sized to form an extension of the tapered surfaces **2508** and **2518**, as well as not block fluid flow through the plug **2500**. The hardened inserts **2510** and **2520** help reduce wear to the plug **2500** over time due to repeated contact of the intake and discharge valves **292** and **294**.

(561) With reference to FIGS. **223-230**, another embodiment of a fluid routing plug **2600** is shown. The fluid routing plug **2600** is generally identical to the fluid routing plug **550**, shown in FIG. **100A-100E**, but the plug **2600** has an alternative embodiment of an intermediate surface **2602**. First and second grooves **2604** and **2606** are formed in the intermediate surface **2602** adjacent the corresponding first and second surfaces **2608** and **2610**. The grooves **2604** and **2606** are configured to receive a first and second seal **2612** and **2614**, as shown in FIGS. **228** and **229**. The seals **2612** and **2614** may be generally identical to the seals **374** and **376**, shown in FIGS. **70** and **71**. However, the seals **2612** and **2614** may have a smaller diameter than the seals **374** and **376**.

(562) Continuing with FIG. **230**, the fluid routing plug **2600** is used with an alternative embodiment of a housing **2616**. The housing **2616** does not include any grooves for housing seals that engage with the fluid routing plug **2600**. Instead, the seals **2612** and **2614** installed within the fluid routing plug **2600** directly engage the walls of the housing **2616**. The area of the housing walls engaged with the seals **2612** and **2614** are characterized as sealing surfaces. In some embodiments, the sealing surfaces may be hardened to help reduce wear and erosion.

(563) In alternative embodiments, a first groove for housing a seal may be formed in the fluid routing plug and a second groove for housing a seal may be formed in the walls of the housing, and vice versa. In such embodiment, a first sealing surface is formed on the fluid routing plug and a

second sealing surface is formed on a wall of the housing, or vice versa.

(564) With reference to FIG. 231, another embodiment of a fluid routing plug **2700** is shown. The fluid routing plug **2700** is a combination of the new features added to the fluid routing plugs **2500** and **2600**. The fluid routing plug **2700** comprises recessed areas **2702** and **2704** for receiving hardened inserts **2706** and **2708**. The fluid routing plug **2700** also comprises first and second grooves **2710** and **2712** for housing first and second seals **2714** and **2716**.

(565) With reference to FIGS. 232-236, another embodiment of a fluid routing plug **2800** is shown. The fluid routing plug **2800** is identical to the fluid routing plug **950** shown in FIGS. 128A-128G, with a few exceptions. A first annular groove **2802** is formed in a first surface **2804** of the plug **2800** for housing a first seal **2806**. Likewise, a second annular groove **2808** is formed in a second surface **2810** of the plug **2800** for housing a second seal **2812**.

(566) Continuing with FIGS. 235 and 236, in operation, the first seal **2806** engages the suction valve **292** and the second seal **2812** engages the discharge valve **294**. The seals **2806** and **2812** create a tight seal between the first and second surfaces **2804** and **2810** and the corresponding valves **292** and **294** during operation. Over time, the seals **2806** and **2812** may be removed and replaced with a new seal, as needed.

(567) With reference to FIGS. 237-248, another embodiment of a fluid routing plug **2900** is shown. The fluid routing plug **2900** is identical to the fluid routing plug **550** shown in FIGS. 100A-100G, with a few exceptions. The fluid routing plug **2900** comprises a plurality of second fluid passages **2902**, each second fluid passage **2902** interconnects opposed first and second surfaces **2904** and **2906** of a body **2908** of the plug **2900**. Like the plug **550**, a counterbore **2910** is formed in the second surface **2906** of the plug **2900**. In contrast to the plug **550**, each of the second fluid passages **2902** does not open directly into the counterbore **2910**. Instead, a group of second fluid passages **2902** open directly into a discharge pocket **2912**, and the pocket **2912** opens directly into the counterbore **2910**.

(568) Continuing with FIGS. 237-248, three second fluid passages **2902** are shown opening into a single pocket **2912**, and four pockets **2912** are shown formed in the body **2908**. The pockets **2912** surround a blind bore **2914** formed at a base **2916** of the counterbore **2910** and have the cross-sectional shape of an arced slot.

(569) In alternative embodiments, more than three second passages **2902** or less than three second passages **2902** may open into a single pocket **2912**. Likewise, more than four or less than four pockets **2912** may be formed within the body **2908**. In further alternative embodiments, the pockets **2912** may have different cross-sectional shapes.

(570) In operation, the pockets **2912** combine the flow of fluid from the second fluid passages **2902** into a single pocket **2912**. The cross-sectional area of a single pocket **2912** is larger than the cross-sectional area of the three corresponding second fluid passages **2902** combined, as shown in FIGS. 255 and 256. The increase in cross-sectional area results in a reduced flow velocity when fluid enters the pocket **2912** from the second fluid passages **2902**. Likewise, the cross-sectional area of the counterbore **2910** is greater than the cross-sectional area of the four pockets **2912** combined, as shown in FIGS. 243 and 244. This further increase in cross-sectional area results in a further reduced flow velocity when fluid enters the counterbore **2910** from the pockets **2912**. The lower fluid velocity reduces erosion of various surfaces within the fluid end caused by fluid flow. The pockets **2912** also help strengthen the body **2908** during operation.

(571) With reference to FIGS. 249-260, another embodiment of a fluid routing plug **3000** is shown. The fluid routing plug **3000** is identical to the fluid routing plug **950** shown in FIGS. 128A-128G, with a few exceptions. The fluid routing plug **3000** comprises a plurality of second fluid passages **3002**, each second fluid passage **3002** interconnects opposed first and second surfaces **3004** and **3006** of a body **3008** of the plug **3000**. Instead of opening directly into a counterbore **3010** formed in the second surface **3006** of the plug **3000**, the second fluid passages **3002** open into a plurality of pockets **3012**. The pockets **3012** are configured identical to and serve the same purpose as the

pockets **2912**, shown in FIGS. **237-248**. In alternative embodiments, the pockets **3012** may be formed like those alternative embodiments contemplated for the pockets **2912**, shown in FIGS. **237-248**.

(572) With reference to FIGS. **261-274**, another embodiment of a fluid routing plug **3100** is shown. The fluid routing plug **3100** is identical to the fluid routing plug **3000** shown in FIGS. **249-260**, with a few exceptions. The fluid routing plug **3100** comprises a plurality of second fluid passages **3102**, each second fluid passage **3102** interconnects opposed first and second surfaces **3104** and **3106** of a body **3108** of the plug **3000**. The second fluid passages **3102** open into a plurality of pockets **3110**. The pockets **3110** open into a counterbore **3112**. The counterbore **3112** comprises a plurality of sidewalls **3114** joined to a base **3116**. The sidewalls **3114** join the base **3116** to a tapered surface **3118** formed in the second surface **3106** of the body **3108**.

(573) In contrast to the fluid routing plug **3000**, the sidewalls **3114** do not have a uniform diameter. Instead, the sidewalls **3114** taper inwardly from the base **3116** to the tapered surface **3118** such that the walls surrounding the counterbore **3112** have a conical shape. Thus, a diameter of the counterbore **3112** is smaller adjacent the tapered surface **3118** than it is adjacent the base **3116**.

(574) Decreasing the diameter of the counterbore **3112** adjacent the tapered surface **3118** increases the surface area of the tapered surface **3118**. Increasing the surface area of the tapered surface **3118** increases the area for the discharge valve **294** to strike and seal against the second surface **3106** of the body **3108**. The increased surface area reduces the strike force applied per unit of area by the valve **294** to the second surface **3106** during operation, thereby increasing the service life of both the valve **294** and the fluid routing plug **3100**.

(575) Continuing with FIGS. **265** and **266**, the fluid routing plug **3100** further comprises a plurality of pockets **3120** formed in the first surface **3104** of the body **3108**. A plurality of the second fluid passages **3102** open into each pocket **3120**. Three second fluid passages **3102** are shown corresponding with a single pocket **3120** in the figures. In alternative embodiments, more or less than three second fluid passages **3102** may correspond with a single pocket **3120**. Four pockets **3120** are shown spaced around a rim **3122** of the first surface **3104** in the figures. In alternative embodiments, more or less than four pockets **3120** may be formed in the first surface **3104**. The pockets **3120** have a cross-sectional shape of an arced slot. In alternative embodiments, the pockets **3120** may have different shapes or sizes.

(576) Continuing with FIGS. **265** and **266**, each pocket **3120** has a greater cross-sectional surface area than the corresponding second fluid passages **3102** combined. In traditional operation, fluid flows into the pockets **3120** and then into the second fluid passages **3102**. However, in some instances, fluid may be caused to flow in a reverse direction. The greater cross-sectional surface area of the pockets **3120** reduces the velocity of fluid flow entering the pockets **3120** from the second fluid passages **3102**.

(577) In operation, a failure of the discharge valve **294** may result in the reverse flow of high-pressure fluid through the second fluid passages **3102** and towards the inlet chamber **106**. The high-pressure fluid may subject any components around the inlet chamber **106**, such as the inlet valve **292** and suction valve guide **296**, to large erosive forces. By allowing the fluid to first enter the pockets **3120** prior to entering the inlet chamber **106**, the velocity of the fluid is reduced, thereby decreasing erosive forces flowing towards the inlet chamber **106** and extending the life of the fluid end.

(578) With reference to FIGS. **275-291**, another embodiment of a fluid routing plug **3200** is shown. The fluid routing plug **3200** comprises a body **3202** having a first surface **3204** joined to a second surface **3206** by an intermediate surface **3208**. The first surface **3204** is also referred to herein as a suction surface **3204**. The second surface **3206** is also referred to herein as a discharge surface **3206**.

(579) Continuing with FIGS. **276**, **278**, and **284-287**, a plurality of first fluid passages **3210** are formed in the body **3202** and interconnect the intermediate surface **3208** and the suction surface

3204. The interconnection may be made by an axially blind bore **3212** formed in the suction surface **3204**. The first fluid passages **3210** are also referred to herein as suction fluid passages **3210**. Three suction fluid passages **3210** are shown in the figures. In alternative embodiments, more or less than three suction fluid passages **3210** may be formed in the body **3202**.

(580) Continuing with FIGS. **286** and **287**, the suction fluid passages **3210** have a generally oval or tear drop cross-sectional shape. An opening **3216** of each suction fluid passage **3210** comprises a first side wall **3218** joined to a second side wall **3220** by first and second ends **3222** and **3224**. The first and second side walls **3218** and **3220** are straight lines of equal length S and the first and second ends **3222** and **3224** are circular arcs, as shown in FIG. **287**.

(581) Continuing with FIG. **287**, the first end **3222** of the opening **3216** has a radius of $R1$ with a center at $C1$, and the second end **3224** has a radius of $R2$ with a center at $C2$. The first end **3222** is larger than the second end **3224** such that $R1 > R2$. The first and second side walls **3218** and **3220** are tangent to the first and second ends **3222** and **3224** and have an included angle, σ . The opening **3216** is oriented such that the center $C1$ of the first end **3222** is a perpendicular distance D from the central longitudinal axis **3214**.

(582) Continuing with FIG. **287**, the opening **3216** has a centerline **3209** that connects the centers $C1$ and $C2$ of the first and second ends **3222** and **3224**. The centerline **3209** has a length E and forms an angle x with the central longitudinal axis **3214**, further orienting the opening **3216**. A cross-sectional shape of each suction fluid passage **3210** through the length of the body **3202** corresponds with the shape of each opening **3216**. The various features of the suction fluid passages **3210** are sized to maximize fluid flow through the suction fluid passages **3210** and minimize turbulence and stress.

(583) With reference to FIGS. **285** and **288** each suction fluid passage **3210** extends between the axially blind bore **3212** and the suction surface **3204** such that each suction fluid passage **3210** comprises a first longitudinal axis **3205** and a second longitudinal axis **3207**. The first longitudinal axis **3205** extends through the center $C1$ of the first end **3222** of the opening **3216**, shown in FIG. **287**. Like the center $C1$, the first longitudinal axis **3205** is offset a perpendicular distance D from the central longitudinal axis **3214**. The first longitudinal axis **3205** does not intersect, is not parallel to, and is not co-planar with the central longitudinal axis **3214**.

(584) The second longitudinal axis **3207** extends through point P , shown on FIG. **285**. The second longitudinal axis **3207** may intersect and be coplanar with the central longitudinal axis **3214**, as shown in FIG. **288**. The configuration of the suction fluid passages **3210** encourages a vortex type fluid flow about the central longitudinal axis **3214** of the body **3202**, thereby reducing fluid turbulence and erosion during operation.

(585) Turning to FIGS. **277-285**, and **288-291**, a plurality of second fluid slots **3226** are formed in the body **3202** and interconnect the suction surface **3204** and the discharge surface **3206**. The second fluid slots **3226** open into a counterbore **3203** formed in the discharge surface **3206**, as shown in FIG. **276**. The second fluid slots **3226** are also referred to herein as discharge fluid slots **3226**. The suction fluid passages **3210** and discharge fluid slots **3226** do not intersect and are in a spaced-relationship with one another. Three discharge fluid slots **3226** are shown in the figures. In alternative embodiments, more or less than three discharge fluid slots **3226** may be formed in the body **3202**.

(586) Continuing with FIGS. **281-285**, and **288-291**, each discharge fluid slot **3226** is generally identical to the group of three second fluid passages **3002**, shown in FIGS. **258-260**, but the walls separating the individual fluid passages **3002** have been removed. Each discharge fluid slot **3226** thus has a larger cross-sectional surface area than the groups of second fluid passages **3002**. Like the group of second fluid passages **3002**, at least a portion of each discharge fluid slot **3226** intersects a corresponding plane **3240** containing the central longitudinal axis **3214**, as shown in FIG. **288**.

(587) Each discharge fluid slot **3226** has an arced cross-sectional shape. The length of the arc may

gradually increase between the suction and discharge surfaces **3204** and **3206**, as shown in FIGS. **277** and **280**. In alternative embodiments, the discharge fluid slots **3226** may have different shapes or sizes.

(588) Continuing with FIGS. **277**, **280**, **284**, each discharge fluid slot **3226** opens on the suction surface **3204** at a first opening **3234** and opens on the discharge surface **3206** at a second opening **3236**. A position of the first and second openings **3234** and **3236** may be determined relative to a plane containing a line **3228** perpendicular to the central longitudinal axis **3214**, as shown in FIG. **284**. The first opening **3234**, when projected onto the plane, is positioned at a first distance **F1** from the central longitudinal axis **3214** and at a first angle $\phi 1$ relative to the line **3228**. The second opening **3236**, when projected on the plane, is positioned at a second distance **F2** from the central longitudinal axis **3214** and at a second angle $\phi 2$ relative to the line **3228**.

(589) The first and second distances **F1** and **F2** shown in FIGS. **277**, **280**, and **284** are different. Likewise, the first and second angles $\phi 1$ and $\phi 2$ shown in FIG. **284** are different. In alternative embodiments, the first and second distances **F1** and **F2** may be the same, but the first and second angles $\phi 1$ and $\phi 2$ may be different. In further alternative embodiments, the first and second angles $\phi 1$ and $\phi 2$ may be the same, but the first and second distances **F1** and **F2** may be different. In even further alternative embodiments, the first and second distances **F1** and **F2** may be the same, and the first and second angles $\phi 1$ and $\phi 2$ may be the same.

(590) The configuration of the discharge fluid slots **3226** encourages a vortex type flow of fluid about the central longitudinal axis **3214** of the body **3202**, thereby reducing fluid turbulence during operation. The shape of the discharge fluid slots **3226** thus maximizes fluid flow and minimizes stress to the fluid routing plug **3200** during operation.

(591) Turning back to FIGS. **275-280**, with the exception of the construction of the suction fluid passages **3210** and the discharge fluid slots **3226**, the intermediate surface **3208** and the suction and discharge surfaces **3204** and **3206** of the fluid routing plug **3200** are identical to those used with the fluid routing plugs **950**, shown in FIGS. **128A-128G**.

(592) With reference to FIG. **292**, another embodiment of a fluid routing plug **3300** is shown. The fluid routing plug **3300** is identical to the fluid routing plug **3200** shown in FIGS. **275-291**, but further comprises a hardened zone **3302** formed in its discharge surface **3304**. The surface of the hardened zone **3302** is harder than its surrounding surfaces and may comprise a wear resistant material, such as tungsten carbide.

(593) A hardened material may be applied to the hardened zone **3302** using a high velocity oxygen fuel (HVOF) thermal spray coating process. Alternatively, the hardened, wear resistant material may be applied to the discharge surface **3304** by other methods known in the art. Following application of the hardened material, the discharge surface **3304** may be ground so that it has a flush surface. Alternatively, the hardened zone **3302** may be finished to a different height, if desired.

(594) In operation, the hardened zone **3302** generally engages the seal **475** installed within the discharge valve **294** as shown in FIG. **89**. The hardened zone **3302** increases the life span of the fluid routing plug **3300** reducing wear and erosion caused by repeated contact between discharge surface **3304** and the discharge valve **294** or seal **475**. In alternative embodiments, the hardened zone **3302** may comprise more surface area of the discharge surface **3304** than is shown in FIG. **292**. In further alternative embodiments, the fluid routing plug **3300** may comprise additional hardened zones, such as a hardened zone on a suction surface **3306** or an intermediate surface **3308** at desired areas.

(595) The housings described herein have various embodiments of suction valves, discharge valves, suction valve guides, and discharge valve guides. One of skill in the art will appreciate that these components may have various shapes and sizes depending on the construction of the housing and various components.

(596) While not shown herein, one of skill in the art will appreciate that the fluid end **100** described

herein may be formed as a single housing having a plurality of horizontal bores formed therein and positioned in a side-by-side relationship. The housing may be attached to a single, large connect plate. In further alternative embodiments, the single housing described above may be broken up into one or more sections having two or more horizontal bores formed therein. Such housings may be attached to one or more connect plates.

(597) One of skill in the art will further appreciate that various features of the fluid routing plugs, housings, and other components described herein may be modified or changed, as desired. While not specifically shown in a figure herein, various features from one or more of the fluid routing plugs described herein may be included in another one of the plugs. Likewise, various features from one or more of the different housings described herein may be included in another one of the housings.

(598) One or more kits may be useful in assembling the fluid end sections **102**. A kit may comprise a plurality of housings described herein, and a plurality of the corresponding fluid routing plugs described herein. The kit may also comprise a plurality of suction valves, discharge valves, suction valve guides, discharge valve guides, and various seals described herein.

(599) The concept of a “kit” is described herein due to the fact that fluid ends are often shipped or provided unassembled by a manufacturer, with the expectation that a customer will use components of the kit to assemble a functional fluid end. Alternatively, some components are replaced during operation. Accordingly, certain embodiments within the present disclosure are described as “kits,” which are unassembled collections of components. The present disclosure also describes and claims assembled apparatuses and systems by way of reference to specified kits, along with a description of how the various kit components are actually coupled to one another to form the apparatus or system.

(600) The term “means for routing fluid” refers to the various fluid routing plugs described herein and structural equivalents thereof. The term “means for regulating fluid flow” refers to the various suction and discharge valves and suction and discharge valve guides described herein and structural equivalents thereof. A “means for pressurizing fluid” refers to the fluid end and the various embodiments of housings and components installed within or attached to the various housings described herein and structural equivalents thereof.

(601) The various features and alternative details of construction of the apparatuses described herein for the practice of the present technology will readily occur to the skilled artisan in view of the foregoing discussion, and it is to be understood that even though numerous characteristics and advantages of various embodiments of the present technology have been set forth in the foregoing description, together with details of the structure and function of various embodiments of the technology, this detailed description is illustrative only, and changes may be made in detail, especially in matters of structure and arrangements of parts within the principles of the present technology to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed. Changes may be made in the construction, operation and arrangement of the various parts, elements, steps and procedures described herein without departing from the spirit and scope of the invention as described in the following claims.

Claims

1. An apparatus, comprising: a fluid routing plug, comprising: a body having an axially blind bore formed therein; a first surface comprising a tapered wall, at least a portion of the tapered wall configured to engage a valve body; an opposed second surface; an outer intermediate surface joining the first and second surfaces; a plurality of first passages joining a plurality of first openings formed in the outer intermediate surface and a plurality of second openings formed in a wall of the body surrounding the axially blind bore; a plurality of second passages joining a plurality of third openings formed in the tapered wall and a plurality of fourth openings formed in

the second surface; and an entrance interconnecting the axially blind bore and the tapered wall.

2. The apparatus of claim 1, further comprising: an axis extending through the body and interconnecting the first and second surfaces; in which the tapered wall is tapered at an angle of 60 degrees relative to the axis.

3. The apparatus of claim 1, further comprising an outer edge formed on the first surface; in which the tapered wall is situated between the outer edge and the entrance.

4. A fluid end, comprising: a fluid end body having a bore formed therein; and the apparatus of claim 1 installed within the bore.

5. The apparatus of claim 1, in which the plurality of first passages are formed offset from and do not intersect with any of the plurality of second passages.

6. An apparatus, comprising: a fluid routing plug configured to be installed within a fluid end, the fluid routing plug comprising: a first end comprising a first tapered sealing surface and a plurality of first openings formed on the first tapered sealing surface; an opposed second end comprising a second tapered sealing surface; an intermediate surface joining the first and second ends; a plurality of first fluid passages formed therein; and a plurality of second fluid passages formed therein, the plurality of second fluid passages interconnecting the first and second ends; in which each of the plurality of second fluid passages is joined to one of the plurality of first openings in a one-to-one relationship.

7. The apparatus of claim 6, in which the fluid routing plug further comprises a plurality of second openings formed on the second end.

8. The apparatus of claim 6, in which the second end of the fluid routing plug further comprises a base, the base comprising a plurality of second openings, each of the second openings joined to one of the plurality of second fluid passages in a one-to-one relationship.

9. The apparatus of claim 8, in which the second end further comprises an outer edge; in which the second tapered sealing surface extends between the outer edge and the base.

10. The apparatus of claim 6, in which the fluid routing plug further comprises an axis extending through the first and second ends; in which the second tapered sealing surface is formed at a non-zero angle relative to the axis.

11. The apparatus of claim 10, in which the non-zero angle is between 1 and 60 degrees.

12. The apparatus of claim 6, in which the first end further comprises an axially blind bore formed therein, the axially blind bore having an entrance.

13. The apparatus of claim 12, in which the first end further comprises an outer edge; in which the first tapered sealing surface is situated between the outer edge and the entrance.

14. The apparatus of claim 6, in which the intermediate surface comprises: a first sealing surface configured to engage a first seal; and a second sealing surface configured to engage a second seal.

15. The apparatus of claim 14, in which the first and second seals are situated within a fluid end housing.

16. A fluid end, comprising: a fluid end housing comprising a rectilinear bore formed therein; the apparatus of claim 14 situated within the rectilinear bore; a first seal that is the first seal of claim 14, the first seal installed within the rectilinear bore; and a second seal that is the second seal of claim 14, the second seal installed within the rectilinear bore.

17. The apparatus of claim 14, in which the intermediate surface further comprises a beveled surface situated between the first sealing surface and the first end of the fluid routing plug.

18. An apparatus, comprising: a fluid routing plug configured to be installed within a fluid end, the fluid routing plug comprising: a first surface, comprising: a first tapered sealing surface having a plurality of first openings formed therein; and an opposed second surface, comprising: a second tapered sealing surface; and a base having a plurality of second openings formed therein; in which the first tapered sealing surface is configured to engage a first valve; and in which the second tapered sealing surface is configured to engage a second valve.

19. The apparatus of claim 18, in which the first valve is a suction valve; and in which the second

valve is a discharge valve.

20. The apparatus of claim 18, in which the fluid routing plug further comprises a plurality of passages formed therein, the plurality of passages joining the plurality of first openings and the plurality of second openings.
